How Much Is the Option to Refinance Worth, and Who Captures the Realized Gains?

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Abstract

The right to refinance makes a 30-year fixed-rate mortgage cheaper than its contract rate suggests: the borrower can swap into a cheaper loan if rates fall and keep the old one if they rise. Investors price this option daily; its value to borrowers, by contrast, has not been assembled as a historical series. Pricing the option from the borrower's side in every month since 1990, I find it lowers expected financing cost by 31 basis points on average and 110 during 2022–2024—largest, as option pricing implies, when rates are high. These estimates assume optimal exercise and so bound from above what borrowers actually capture; alternative specifications for mortgage-spread dynamics imply a somewhat larger 39–48 basis points. Ex post, exercising the option as rates fell delivered households roughly \$3.3 trillion in present-value gains, in constant 2026 dollars, over 1990–2024 (\$2.1–\$2.6 trillion net of what the option was worth at origination, depending on the cost basis)—a windfall that appears in no income or wealth statistic, that payment-based measures of refinancing savings conflate with cash-outs and stretched-out repayment, and that is the mirror image of investors' prepayment losses. The payoff was overwhelmingly luck rather than predictable premium, and it was shared unevenly: per dollar of mortgage debt held, Black borrowers captured 0.59 of a proportional share and the bottom income quintile 0.82, versus near parity for white and top-quintile borrowers. A further \$2.2 trillion of locked-in below-market mortgages remained outstanding in 2022.

JEL Codes: G21, G12, D14, D31, E52, R31
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1 Introduction

The right to refinance a 30-year fixed-rate mortgage is one of the most valuable financial options most households will ever hold: if rates fall the borrower can swap into a cheaper loan, and if they rise the borrower simply keeps the old one. Investors who buy mortgages price this option carefully, because every refinance costs them a high-paying loan, yet borrowers are never shown its value. The rate quoted at origination—the number that sets the payment and decides who qualifies—is the cost of a loan held unchanged for thirty years. For a borrower who will refinance when it pays to, that number overstates the loan's true expected cost, though by how much has not been quantified for borrowers. The machinery that could quantify it exists on Wall Street, where investors who own mortgages calculate every day what borrower refinancing will cost them; but those models are proprietary, and they price the investor's loss, not the borrower's expected cost of financing a home.

The same option, exercised across three decades of generally falling rates, also produced one of the largest windfalls American households have ever received—and one of the least visible. Refinancing reduced what households owed on their mortgages, measured in constant 2026 dollars, by roughly $3.3 trillion between 1990 and 2024. Almost none of that gain was ever counted. It is not income: trading an expensive loan for a cheap one puts no cash in anyone's pocket. It is not measured wealth: the loan balance is nearly unchanged the day after a refinance, so the household's balance sheet looks the same even though what it must pay over the coming decades has fallen. And research on refinancing typically measures something else entirely—the change in the monthly payment—which mixes the true gain together with cash taken out at closing and with the relief that comes from stretching repayment back over a fresh thirty years. In 2020, only about $88 of the average $233 drop in monthly payments came from lower rates; the rest came from stretching out repayment. The windfall is real and enormous, but it has gone missing from the statistics—and from research that measures refinancing by its effect on monthly payments.

This paper asks two questions. First, how much is the right to refinance worth at origination—equivalently, what is a fixed-rate mortgage expected to cost? Second, how much did households actually gain by exercising that right as rates fell, and who captured those gains?

For the first, I compute the steady mortgage rate that would leave a borrower as well off, over the life of the loan, as the quoted rate paired with the right to refinance—call it the option-adjusted cost rate; the gap between it and the quoted rate is what the option is worth. This needs only a rule for when to refinance, a model of interest rates anchored to the conditions prevailing at origination, and a simulation over thousands of future rate paths, repeated for every month from 1990 through early 2026. For the second, I treat the mortgage as a debt the household can retire early at face value. When rates fall the loan is worth more than its balance, but the borrower pays it off at the balance by refinancing; the difference, net of costs, is a capital gain. Measuring that gain requires stripping out the part of any payment cut that merely stretches repayment back to thirty years; I track separately the refinances in which the borrower also drew out cash. The distributional question—who earned those gains—is answered with the near-complete record of mortgages collected under federal lending-disclosure law (the Home Mortgage Disclosure Act), loan-level histories from Freddie Mac, and the Survey of Consumer Finances.

Together the two questions deliver all three sides of one option contract: what it is worth, what the market charged for it, and what it paid off. Averaged over thirty-six years, the option is worth about 31 basis points—roughly a third of a percentage point off the quoted rate—and its value swings with the rate environment, rising past a full percentage point during the 2022–2024 rate spike. The option is most valuable in exactly the environments where borrowers feel most squeezed by high rates. And the market has generally charged more than that: the part of the mortgage rate that investors attribute to the risk of refinancing exceeds what the option is worth even to a borrower who refinances perfectly, consistent with the extra compensation investors require for bearing prepayment risk.

The payoff dwarfed both the price and the value. As rates trended down over 1990–

2024, households captured roughly \$3.3 trillion in gains, in constant 2026 dollars—about \$2.6 trillion more than the option was worth when their loans were made, and at least \$2.1 trillion counting only refinances that swapped the rate without taking cash out. Most of that excess was luck: rates fell farther than markets or forecasters expected, so the gain was a surprise.

The gains were also shared unevenly. Higher-income households collected most of the dollars largely because they hold most of the mortgage debt; the racial gap, by contrast, survives that adjustment—per dollar of mortgage debt, Black borrowers captured 0.59 of a proportional share, against near parity for white borrowers. What households gained, mortgage investors lost: a vast, uneven transfer that no income or wealth statistic records. And the flip side of the same option appeared when rates rose: by 2022 households were sitting on \$2.2 trillion of mortgages cheaper than any loan they could newly get, the "lock-in" that has since reshaped the housing market.

The investor's side of this option has been modeled for decades; this paper builds the borrower's-side counterpart: a monthly series, spanning thirty-six years, of what the right to refinance is worth to the borrower at origination, anchored each month to the interest rates prevailing at the time, paired with an accounting of the gains the option actually delivered and the record of who captured them. The estimates assume a borrower who refinances at the right moments, so they are a ceiling on what real borrowers, who often fail to act, capture. And the option-adjusted cost rate is an expected long-run financing cost, not a promise that any given mortgage is affordable.

1.1 Related Literature

The paper draws on three literatures and fills a gap in each.

The first studies the borrower's side of the mortgage—the canonical problem of household finance (Campbell, 2006). Agarwal et al. (2013) solve the borrower's refinancing problem in closed form. A large empirical literature shows that the resulting option is exercised imperfectly: roughly 20% of unconstrained borrowers fail to refinance when it is optimal

(Keys et al., 2016), deeply in-the-money options go unexercised (Deng and Quigley, 2001), inaction is widespread and tied to inattention and complexity (Andersen et al., 2020; Berger et al., 2024b), and documentation requirements and out-of-pocket costs block refinancing precisely when it is most valuable (DeFusco and Mondragon, 2020). Gerardi et al. (2023) show that Black and Hispanic homeowners are significantly less likely to refinance, amplifying the redistributive effects of monetary policy, and Berger et al. (2024a) and Fisher et al. (2024) study how refinancing frictions redistribute wealth across borrowers with heterogeneous rates. This literature measures how the option is used and misused; how much it is worth at origination, and what its exercise actually delivered in dollars, is left open.

The second values the prepayment option from the investor's side. Contingent-claims models of mortgage-backed securities (Dunn and McConnell, 1981; Schwartz and Torous, 1989; Stanton, 1995; Deng et al., 2000) price the borrower's right to prepay as the investor's risk, and Gabaix et al. (2007) and Boyarchenko et al. (2019) show that bearing that risk commands a premium in MBS spreads. Related work on product choice and monetary transmission—the fixed-versus-adjustable decision (Campbell and Cocco, 2003; Koijen et al., 2009) and the refinancing channel of monetary policy (Berger et al., 2021; Eichenbaum et al., 2022)—establishes why the option matters for household choices and for policy. These models price the investor's loss, not the borrower's expected cost of financing a home, and none translates the option into a borrower-side value at origination.

The third measures what borrowers gain from refinancing, almost always in cash-flow terms: the change in the monthly payment or the discounted stream of payment differences (Hurst and Stafford, 2004; Keys et al., 2016; Andersen et al., 2020), or the cash-flow relief refinancing delivers to constrained households (Beraja et al., 2019). For consumption stimulus the payment change is the relevant measure. As a measure of the wealth transferred by exercising the option it is not, because a payment can fall when the rate falls, when the term resets to thirty years, or when the borrower extracts equity at closing, and only the first is a gain in household wealth. Section 4 develops a present-value measure that isolates that component; to my knowledge, the resulting series—the realized wealth gain from refinancing, aggregated over 1990–2024 and allocated across households—has no published counterpart.

The contribution is to measure, for the same option contract and from the borrower's vantage, what these literatures treat separately: the option's value at origination, month by month over 1990–2026; the market's charge for the same risk; and the realized payoff and its distribution. The behavioral evidence above makes the optimal-exercise benchmark more useful, not less: if exercise is imperfect (Keys et al., 2016), unequal (Gerardi et al., 2023), and hampered even at origination (Bhutta et al., 2026), the benchmark prices the stakes of those frictions.

The paper proceeds as follows. Section 2 fixes the key quantities and the accounting identity that links them, summarizes the measurement approach, and describes the data. Section 3 presents the ex-ante option-value series, the reasons it varies, and its relationship to MBS pricing. Section 4 measures the gains households realized and the surprise component; Section 5 measures who captured them. Section 6 concludes with implications. The appendix collects the formal model, derivations, robustness checks, and supplementary results.

2 Framework, Measurement, and Data

This section fixes the quantities the paper measures and the accounting that links them, so that the ex-ante value of the refinancing option and its ex-post payoff are never conflated; it then summarizes how the option is valued and describes the data.

2.1 The mortgage as a callable liability

A fixed-rate mortgage is an amortizing callable liability. The borrower is short a non-prepayable amortizing mortgage bond and long an American-style call to repurchase that liability at its scheduled unpaid balance; a refinance exercises this call, financed by issuing a new prepayable loan. When market rates fall, the market value of the borrower's above-market liability rises above par, but the borrower extinguishes it at par—and the difference, net of costs, is a realized capital gain on the liability side of the household balance sheet. The option the borrower holds is the prepayment risk that mortgage investors are short: the borrower's gain is the investor's loss, so the aggregate payoff is a redistribution rather than new wealth.

2.2 An accounting identity

The contract rate is the payment-relevant quote, but it overstates the borrower's expected financing cost because it prices the loan as if held unchanged for its full term. It can be decomposed as

$$\underbrace{m(0)}_{\text{contract rate}} = \underbrace{c^*}_{\text{option-adjusted cost rate}} + \underbrace{V}_{\text{option value}}, \tag{1}$$

where c^* is the constant rate that delivers the same expected present-value cost on a hypothetical no-option mortgage and $V = m(0) - c^*$ is the value the borrower derives from the refinancing option. Equation (1) is a present-value equivalence identity, not a structural decomposition of the mortgage rate. Wedges common to the callable and no-option counterfactual—credit risk, servicing, and the primary–secondary spread—cancel when held fixed; future variation in the all-in mortgage spread, however, affects the value of refinancing and is included in V (Appendix Table 12 reports the sensitivity to alternative spread processes). Investors price the same risk continuously; this paper measures the option's value from the borrower's side ex ante (Section 3) and then measures the gain the option actually delivered ex post (Section 4).

2.3 Price, payoff, and surprise

I keep three quantities distinct. The ex-ante option value V is what the option is worth at origination under optimal exercise; the market's own charge for the same risk—the prepayment premium in MBS spreads—can exceed it, and the two are compared in Section 3.3. The realized refinancing payoff P^{realized} is the gross gain households actually captured by exercising as rates fell over 1990–2024; net of the dollar present value of that ex-ante premium (C_{model} in Table 1) it is the net realized gain. The surprise is the gap between the value realized under optimal exercise on the actual rate path and the ex-ante value; a decomposition against survey-expectations and random-walk drift benchmarks (Appendix N) shows that only a small part of it was predictable from the term-premium wedge between forward and expected rates, with the bulk residual path luck. I lead with the gross payoff and report the net gain as a second, Haig–Simons framing, since the premium is a model-implied value rather than an observed cash basis. A distinct lock-in stock—the unrealized below-market value of outstanding mortgages when rates rise—is a current mark, never added to the realized flow. Table 1 collects these quantities and their baseline values; the rest of the paper estimates each in turn.

Table 1: Key quantities and baseline values

Quantity	Symbol	Definition	Baseline value
Contract rate	$m(0)$	The quoted 30-year fixed rate; sets the monthly payment and qualification.	data
Option-adjusted cost rate	c^*	Constant no-option rate giving the same expected present-value cost.	$m(0) - 31$ bps (avg)
Ex-ante option value	$V = m(0) - c^*$	What the refinancing option is worth at origination under optimal exercise.	31 bps avg; 143 bps max
Realized refinancing payoff (gross)	P^{realized}	Present-value liability reduction, net of cost, captured by households that refinanced (rate reduction on all existing balances refinanced), 1990–2024.	$\sim$ \$3.3 trillion
Net realized gain	$P^{\text{realized}} - C_{\text{model}}$	Gross payoff net of C_{model}, the dollar present value of the ex-ante option premium (a Haig–Simons capital gain).	$\sim$ \$2.1–\$2.6 trillion
Surprise	realized optimal $-V$	Value realized under optimal exercise on the actual path, minus the ex-ante value V; mostly path luck, with a small predictable term-premium wedge (App. N).	44 bps $\rightarrow$ 89 bps
Lock-in stock	$B - L^{\text{callable}}$	Unrealized below-market value of outstanding mortgages when rates rise; a current mark, never summed into the flow.	$\sim$ \$2.2 trillion (2022)

2.4 Measurement in Brief

Computing the option-adjusted cost rate requires a decision rule for when to refinance, a model of future interest rates consistent with conditions at origination, and an average over the many paths rates might follow. The analysis combines three components: the closed-form optimal refinancing threshold of Agarwal et al. (2013), a Hull–White one-factor term-structure model (Hull and White, 1990) recalibrated each month to the observed Treasury yield curve, and a Monte Carlo simulation that applies the decision rule along 10,000 simulated rate paths per origination month. This subsection describes what each component does; the formal model, the calibration, and the validation and convergence checks are collected in Appendices B–K.

The decision rule solves a nontrivial stopping problem: a borrower should not refinance as soon as the present-value savings cover the closing costs, because exercising the option today forecloses exercising it at an even better rate tomorrow. Agarwal et al. (2013) derive the optimal rule in closed form—refinance when the gap between the current rate and the market rate exceeds a threshold that rises with rate volatility and closing costs and falls with the expected time remaining in the loan. The threshold is far from the naive break-even: at their baseline parameters it is 139 basis points, more than three times the naive 44-basis-point level, and under the January 2024 calibration used here it is 91 bps (Appendix B states the formula and replicates their published thresholds).

Rate dynamics come from the Hull–White model, calibrated each month so that simulated paths reprice the Treasury yield curve observed on that date and match the trailing volatility of 10-year yields. Simulated Treasury paths are converted to mortgage-rate paths using an estimated pass-through coefficient on monthly Treasury changes (about 0.6, estimated on expanding windows to avoid look-ahead bias) plus a persistent stochastic mortgage–Treasury spread, with each path anchored at the observed contract rate; Appendix C states the equations, and the spread specification—the largest source of model uncertainty—is varied in Appendix Table 12. The resulting series is a forward-curve-consistent borrower-side benchmark: the expected financing cost under rate dynamics whose drift matches the term structure observed at origination and whose volatility matches trailing history, discounted at the observable 10-year Treasury yield—the market opportunity cost of funds rather than an unobserved preference parameter, an approach standard in borrower-side mortgage valuation (Stanton, 1995). Because the forward curve embeds term premia, the drift is not a pure physical forecast; replacing it with survey-expected rates changes the mean ex-ante value by only 3 bps over the cohorts where the comparison is feasible (Appendix N), so the choice of drift is quantitatively minor for the headline series.

The simulation applies the threshold along each path: when it is crossed, the borrower's rate resets to the market rate, she pays that era's closing costs, and the loan re-amortizes over a fresh thirty years. Cash flows are weighted by the probability that the mortgage is still outstanding, so the effective horizon is the borrower's expected loan life—about 8.7 years in the early 1990s, rising to 12.3 years in the 2020s as homeowner mobility fell—rather than an arbitrary fixed cutoff. The option-adjusted cost rate c^* is the constant rate on a hypothetical no-option mortgage that delivers the same expected present-value cost, inclusive of refinancing costs and terminal balances, as the contract rate paired with optimal refinancing; the option value is $V = m(0) - c^*$, in basis points. The formal present-value-equivalence condition is stated in Appendix F (Equation (9)). Each monthly estimate uses only information available at the origination date, and the Monte Carlo standard error is about ± 1 basis point at 10,000 paths (Appendix H).

2.5 Data and Calibration

Interest-rate data come from the Federal Reserve Economic Data (FRED) database: the Freddie Mac Primary Mortgage Market Survey 30-year fixed rate (PMMS, series `MORTGAGE30US`) and Treasury constant-maturity yields at 11 maturities from 1-month through 30-year. The sample takes the first business day of each month from January 1990 through February 2026—434 monthly observations—and extracts the contract rate, the full yield curve, the 10-year-minus-2-year slope, trailing 36-month mortgage-rate volatility, and the mortgage–Treasury spread. The mortgage rate ranges from 2.67% (January 2021) to 10.56% (May 1990); replication code and data are described in Appendix Q.

The four borrower-side parameters of the refinancing threshold—the mortgage balance, the effective tax rate on mortgage interest, the annual moving probability, and origination costs—are calibrated to time-varying public data rather than fixed at representative-borrower values, so that thresholds and option values reflect the conditions borrowers actually faced at origination. The average GSE loan balance rose from roughly \$102,000 in 1990 to \$350,000 in 2024; origination points fell from about 2% to 0.5–0.7%; homeowner mobility fell from about 9% to 4–5%, lengthening the survival-weighted expected loan life; and the effective tax rate on mortgage interest fell from roughly 0.15–0.18 to about 0.065 after the 2018 doubling of the standard deduction. Larger balances and cheaper originations lower the threshold and raise the option value; longer expected tenure raises it; the tax change works modestly in the opposite direction. Appendix I documents each series and its sources. Appendix Table 16 shows that the time-varying calibration changes the full-sample mean only modestly relative to a fixed representative borrower (25 versus 27 bps, holding the discount rate and horizon fixed), though it shifts the decade pattern; the observable real-Treasury discount and the survival-weighted horizon then raise the mean to 31 bps.

2.6 Data for the Realized-Gains Analysis

The ex-post analysis of Section 4 draws on three additional public sources. The distribution of realized refinancing flows comes from the Home Mortgage Disclosure Act (HMDA) loan-level records, which cover essentially the universe of mortgage originations: I use the pre-2018 Loan/Application Register files (harmonized across the 2004 change in race and ethnicity coding) together with the post-2018 modern files, which additionally report the contract rate, for every year 1990–2024. Aggregate refinance volumes are taken from the Mortgage Bankers Association and validated against HMDA.

The mechanics of take-up—the probability that an in-the-money borrower actually refinances—are estimated from the Freddie Mac single-family loan-level dataset, a loan-month panel of repayment and prepayment outcomes (about three million sampled loan-months). The household balance-sheet interpretation—the wealth rank and birth cohort of refinancing households, the book-versus-market value of their mortgages, and the unrealized lock-in stock—comes from the Federal Reserve's Survey of Consumer Finances (SCF). Appendix A documents each source, its coverage, and the imputations.

3 The Value of the Refinancing Option, 1990–2026

3.1 The Option-Adjusted Cost Rate Series

For each of the 434 months from January 1990 through February 2026, I calibrate the model to that date's yield curve, contract rate, trailing volatility, and borrower parameters (Section 2.4), and compute the option-adjusted cost rate and option value as cross-path averages. Each monthly estimate reflects only information available at the origination date—no future rate movements are used.

Figure 1 presents the main finding. For each month in the sample, I plot the 30-year fixed mortgage contract rate alongside the option-adjusted cost rate.

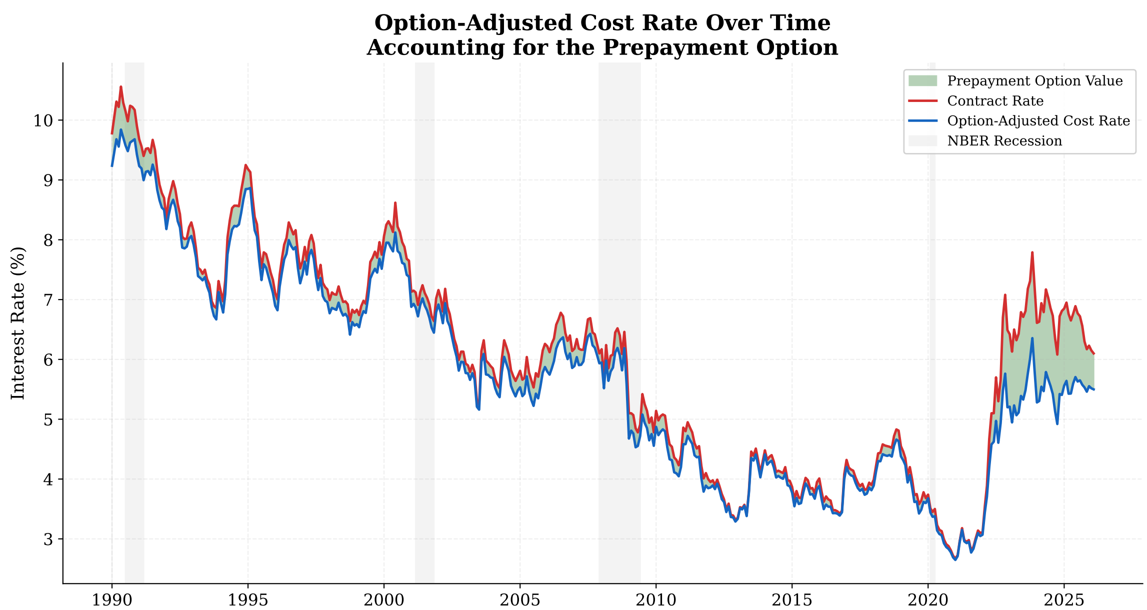

Figure 1: The red line shows the 30-year fixed mortgage contract rate. The blue line shows the option-adjusted cost rate, accounting for optimal refinancing over the survival-weighted expected loan life (10,000 paths per month). The green shaded area between the two lines is the refinancing option value. Gray bars indicate NBER recessions.

The option-adjusted cost rate (Appendix F, Equation (9)) is at or below the contract rate in every month, but the gap is state-dependent. Across the full sample, the option value averages 31 bps with a median of 22 bps (Table 2). The distribution is right-skewed: the 90th percentile is 65 bps, and the maximum is 143 bps (November 2023, when the contract rate was 7.8%). In elevated-rate environments (contract rate above 6%), the option value is substantially larger.

Table 2: Option Value Summary Statistics by Period (basis points)

	Full Sample	1990s	2000s	2010s	2020s	2022–24
Mean	31	28	26	12	74	110
Median	22	24	26	10	75	130
90th pct	65	45	37	22	138	140
Max	143	72	50	28	143	143
N (months)	434	120	120	120	74	36

Notes: Option value is $V = m(0) - c^*$ in basis points, where $m(0)$ is the contract rate and c^* is the option-adjusted cost rate (PV-equivalent measure; see Appendix F). Computed from 10,000 Monte Carlo paths per month over the survival-weighted expected loan life (Appendix F). Decades are defined by origination date (1990s = Jan 1990–Dec 1999, etc.); the 2022–24 column is a subset of the 2020s. Borrower parameters (M, τ, μ, points) are time-varying as described in Section 2.5; remaining parameters as in Appendix J.

The interquartile band is wide, reflecting substantial heterogeneity across simulated paths. In some paths the borrower never refinances and pays the full contract rate; in others, she refinances multiple times and achieves a rate well below the mean.

3.2 Why the Option Value Varies

Two complementary views of the option-value distribution across the full sample—box plots by decade and a scatter of option value against the contract rate, colored by yield-curve slope—appear in Appendix P (Appendix Figure 15).

The 2020s exhibit the highest option values (mean 74 bps), reflecting elevated rates, high volatility, and longer expected borrower tenure. The 2000s and 1990s show moderate values (means of 26 and 28 bps), while the 2010s—a prolonged low-rate, low-volatility era—has the lowest average (12 bps).

The option value depends on three features of the rate environment. Rate volatility is the strongest driver: higher volatility generates more dispersion in future rate paths, creating more refinancing opportunities—the standard result from option pricing that volatility benefits the option holder. The contract rate level matters because a higher starting point provides more room for rates to fall, though this effect is attenuated because high-rate decades also featured smaller loan balances and higher origination costs that raised the refinancing threshold (Section 2.5). The yield curve slope enters negatively: conditional on rate level and volatility, a steeper curve signals expected rate increases, reducing the probability that rates will cross the refinancing threshold.

These dependencies are summarized in a heatmap of the mean option value by mortgage-rate level and yield-curve slope (Appendix P, Appendix Figure 16): the highest values cluster where high rates combine with moderate or flat yield curves, a pattern present in the early 1990s and again during 2022–2024.

Two dates make the state dependence concrete. A borrower originating in January 2024 at 6.61%—elevated volatility, a slightly inverted yield curve, and an optimal threshold of 91 bps—expects to refinance about 2.8 times over the loan's expected life, for an option value of 133 bps, or roughly $29,000 in present value on that year's average balance. The same borrower in January 2021 at 2.67%, with volatility near its sample low, expects to refinance 0.17 times, and the option is worth about 2 bps: 86% of simulated paths never refinance in that environment, against 3% in January 2024. The option is worth the most precisely when rates are highest and borrowers feel most burdened by their mortgage costs. Appendix M reports the full calibration values for these scenarios; Appendix P adds a by-regime summary (Appendix Table 17), sample simulated rate paths, and the refinancing-event distributions (Appendix Table 18).

A historical backtest (Appendix N) shows that months with higher predicted option values do experience greater realized savings (correlation 0.61 across $N = 314$ cohorts with full 10-year windows)—ranking power across origination environments rather than point-prediction accuracy. The backtest also systematically under-predicts the savings actually realized; that gap is the realized "surprise" analyzed in Section 4.

3.3 The Market's Price of the Same Option

A natural external comparison sets the borrower-side option value alongside how mortgage investors price the same prepayment risk. Investors in mortgage-backed securities demand a spread over Treasuries to compensate for the chance that borrowers refinance, and the prepayment-related part of that spread—the MBS spread net of the option-adjusted spread—is the market's own valuation of the option. Figure 2 plots this market option cost, extracted from Bloomberg current-coupon yields, against the model's option value. The two series co-move strongly and positively—the secondary-market spread rises 0.56 bps per basis point of option value (Newey–West $t = 4.94$)—and the relationship runs specifically through the secondary-market (prepayment) spread rather than the primary–secondary spread, consistent with investors pricing prepayment risk specifically. The borrower-side benchmark and the investor-side price thus reflect similar underlying economics, without requiring a literal exercise-probability interpretation. Because both series are persistent, I check that this is not a spurious level correlation: the secondary spread and the option value are cointegrated, with an error-correction term of -0.20 per month (Newey–West $t = -4.88$) pulling the spread back toward its option-value-implied level, while short-run monthly changes are uninformative as expected. Appendix O reports the construction, the full specification table, and the caveats.

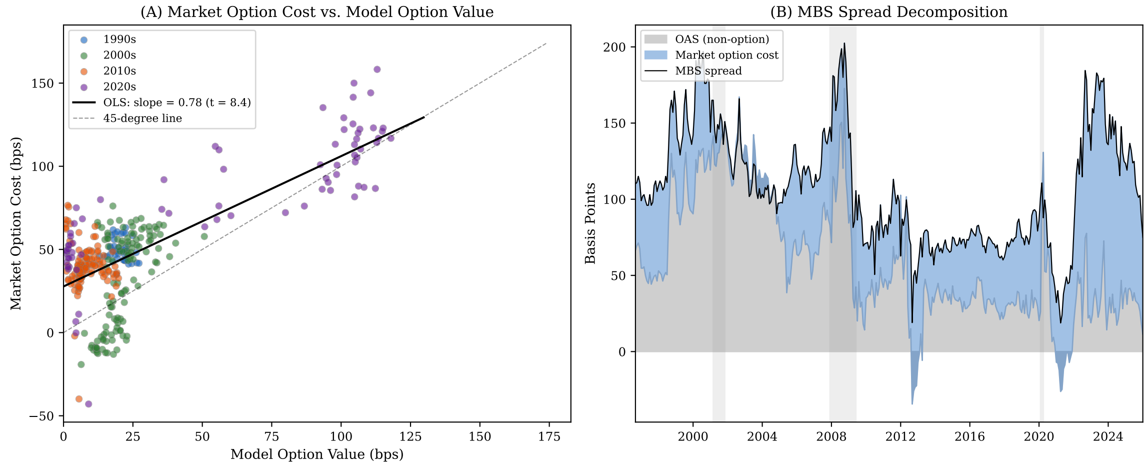

Figure 2: Borrower-side option value and MBS pricing. Panel A: market option cost (MBS spread minus OAS) vs. calibrated model option value for 353 monthly observations (September 1996–January 2026), colored by decade, with OLS regression line and 45-degree line. The slope printed in Panel A is for the market-option-cost regression; the preferred slope quoted in the text (0.56) is for the secondary-market spread—Appendix Table 15 reports every specification. Panel B: time-series decomposition of the MBS spread into OAS (gray, non-option component) and market option cost (blue, prepayment-related component).

The two series are the same option seen from opposite sides of the market, and they differ for identifiable reasons. They agree because both price the borrower's right to prepay a 30-year fixed-rate loan: when rates are high and volatile, the model says the option is worth more to the borrower, and investors simultaneously demand more compensation for the risk that it will be exercised against them. They differ because the market's charge embeds more than the option's optimal-exercise value: it prices actual (imperfect and heterogeneous) exercise behavior rather than the optimal rule, it is set under the risk-neutral measure while the model works under physical dynamics, and it includes the premium investors require for bearing prepayment risk at all. In most years the market charge exceeds the model's optimal-exercise value, consistent with the prepayment-risk premium that Gabaix et al. (2007) and Boyarchenko et al. (2019) document from the investor side. Over the full sample the option value averages roughly a quarter of the secondary-market spread, from about a tenth in the 2010s to over half in the 2020s. The wedge between the market charge and the optimal-exercise value is also what separates the two net-gain bases in the realized-gains accounting (Section 4, Appendix Table 1).

3.4 Robustness

The option value behaves as option pricing predicts: it rises with rate volatility—the strongest driver—and with the rate level, balance, and expected horizon, and it falls with closing costs and with exercise frictions. All robustness evidence is collected in Appendix L, indexed by Appendix Table 11: one-at-a-time sensitivity curves (Appendix Figure 9); the spread-model variants—the largest source of model uncertainty, spanning 31–48 bps under the headline configuration with cross-month correlations of 0.92 or higher (Appendix Table 12); and the discount and horizon assumptions, which bracket the headline between 22 and 32 bps (Appendix Table 13). The rate-model, threshold, volatility, and discretization checks (Appendix K) each move the value by only a few basis points.

4 The Realized Gains from Refinancing, 1990–2024

The option-adjusted cost rate of Section 3 prices the refinancing option ex ante. This section turns to the ex post side—how much borrowers actually gained by exercising that option as rates fell over 1990–2024, why the gain so far exceeded the price, and why it appears in no income or wealth statistic. Section 5 then measures who captured it. The quantities and their baseline values were defined in Section 2 (Table 1); here I estimate the ex-post ones.

4.1 Measuring the realized payoff

Building on the callable-liability framing of Section 2.1, I measure the realized gain with a terminal-balance-adjusted present value (defined in Appendix A) that prices the remaining old and new cash flows and both terminal balances, so that re-amortizing a seasoned loan back to thirty years is not rewarded as wealth.

> A concrete transaction. A household with a $250,000 balance at 10% refinances into 2%. The scheduled payment falls from about $2,194 to $924 per month—a raw saving

near \$1,270 per month. But much of that relief is liquidity, not wealth: resetting a seasoned loan to a fresh thirty-year term lowers the payment by stretching repayment, which the terminal-balance adjustment removes. The wealth gain is the present value of the lower financing cost on the balance—about \$55,000 here (roughly 22 cents per dollar refinanced)—not the capitalized payment reduction. Empirically the distinction is first-order: in 2020 only about \$88 of the average \$233 monthly payment reduction was the rate effect; the remaining \$145 was re-amortization.

Aggregated to a national flow, the realized payoff in year t is $G_t = \max(g_t, 0) \times V_t^{\text{exist}}$, where g_t is the per-dollar payoff (eq. (2) in Appendix A) and $V_t^{\text{exist}} = V_t^{\text{rt}} + (1 - q_t)\, V_t^{\text{co}}$ is existing-loan refinance volume—rate-and-term loans in full plus the old-balance portion of cash-out refinances (eq. (3) in Appendix A). The cumulative flow, $P^{\text{realized}} = \sum_t G_t$, is the gross payoff of Table 1; the net realized gain is the payoff net of the ex-ante option value V (Section 2). This is a liability-side capital gain that standard statistics do not record as a realized gain or a liability revaluation: it generates no cash receipt, so it does not enter measured income as an option-exercise payoff, and it leaves the unpaid principal little changed, so it does not enter book net worth in the Survey of Consumer Finances or the Distributional Financial Accounts. Its effects appear only indirectly—through lower future interest payments, higher saving, or higher consumption.

4.2 Aggregate realized gains, 1990–2024

Figure 3 plots the annual and cumulative realized payoff. The per-dollar gain is large in every refinancing wave (about 16 cents per dollar in 1992–93, 7 cents in 2001–03, 8–9 cents in 2009–13 and 2020–21), and the cumulative gross realized payoff since 1990 is about \$3.3 trillion: the rate reduction realized on every existing balance refinanced—rate-and-term loans in full plus the old-balance portion of cash-out refinances (treated below). For scale, that cumulative payoff is roughly a quarter of the \$13.4 trillion of one-to-four-family mortgage debt outstanding at the end of 2024 (FRED series `HHMSDODNS`). The wave structure is summarized in Table 3: on the headline existing-loan basis, the 2009–13 HARP-era refinancings delivered roughly $600 billion gross and the 2020–21 pandemic boom about $370 billion, the two largest episodes. Of the $3.3 trillion gross, about $474 billion comes from the modeled pre-1999 years; the remainder is measured from observed refinance volume and, from 1999 on, loan-level rates.

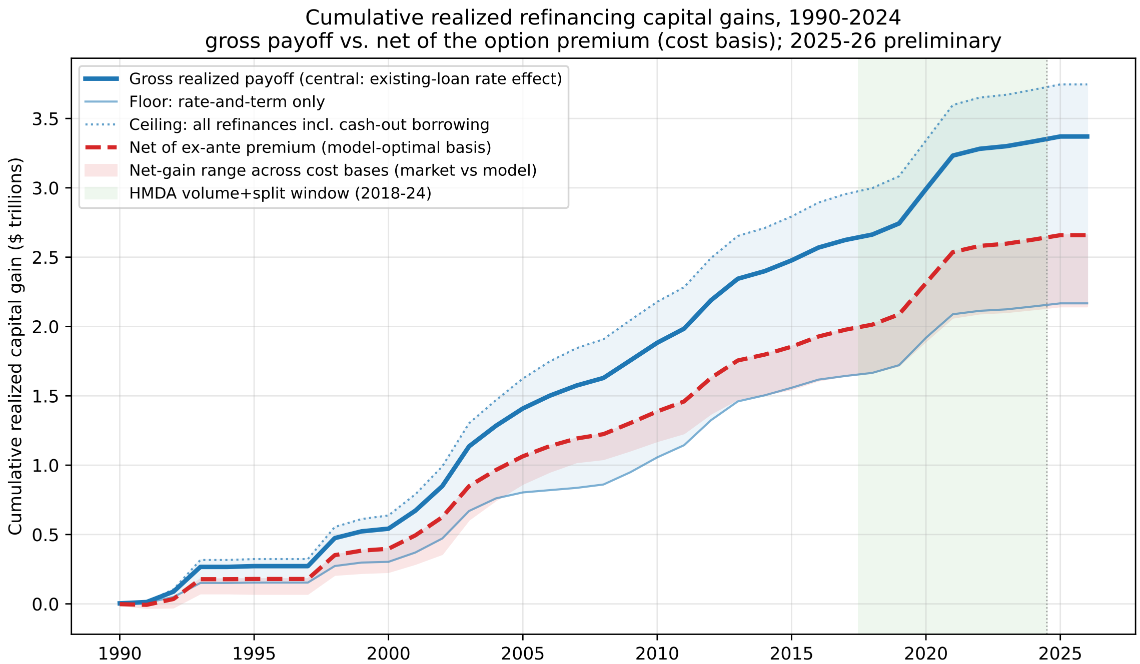

Figure 3: Cumulative realized refinancing payoff, 1990–2024. The central estimate (solid) applies the realized per-dollar rate gain to all existing balances refinanced—rate-and-term loans in full plus the old-balance portion of cash-out refinances—reaching about $3.3 trillion gross and $2.6 trillion net of the ex-ante option value (dashed; the thin band spans the $2.1–$2.6 trillion range across cost-basis assumptions, Appendix Table 1). The shaded band runs from the rate-and-term floor ($2.1 trillion, cash-out excluded) to the ceiling that also credits the cash-out proceeds ($3.7 trillion). Per-dollar gains are loan-level actual (Freddie, 1999+) and a median-age-calibrated model before; volume is MBA, validated against HMDA. All amounts are in constant 2026 dollars (PCE deflator); 2025–26 preliminary.

Table 3: Realized refinancing capital gain by episode (headline series), in constant 2026 dollars. Volume and rates are rate-and-term refinances; the source column flags the data tier.

Episode	Volume	Share	Old rate	New rate	Gain/$	Gross	Net	Source
1992-93	455	0.476	9.92	7.63	0.1559	143	104	Modeled
1998	445	0.516	9.15	6.95	0.1461	119	102	Modeled
2001-03	3063	0.562	7.05	6.14	0.0708	367	280	GSE + MBA
2009-13	4774	0.805	6.41	4.38	0.0896	599	448	GSE + MBA
2020-21	3718	0.719	3.68	2.96	0.0808	367	337	GSE + HMDA

Notes: Volume = V^{rt}, rate-and-term refinance volume ($bn); Share = rate-and-term share of total refinance volume; Old/New rate = volume-weighted average coupon rates (%); Gain/$ = g_t, the per-dollar realized gain (eq. (2)); Gross and Net = G_t and G_t^{net} cumulated over the episode ($bn, eqs. (3)–(4)), on the headline existing-loan basis. Source: Modeled = PMMS-lag model gain with MBA volume (pre-1999); GSE + MBA = loan-level per-dollar gain with MBA volume; GSE + HMDA = loan-level gain with HMDA-era volume split. Generated by 08_realized_gains_figures.py.

Two presentational points matter. First, the per-dollar series must be read as two distinct measures (Appendix Figure 1, Appendix A.2): a representative opportunity value for the average outstanding loan, which turns negative in lock-in years (2022–24) when the typical borrower is out of the money, and the realized gain among refinancers, which stays positive throughout because the households that refinance are positively selected into being in the money. The dollar aggregate uses the latter, so a lock-in environment never appears as a negative realized gain. Second, cash-out refinances belong in the total, but only their existing-loan component. A cash-out refinance cuts the rate on the old balance (a genuine refinancing gain), while the cash-out proceeds are new borrowing at the new rate (no gain). The central estimate therefore applies the realized rate gain to all existing balances refinanced—rate-and-term volume in full, plus each cash-out loan's old balance, imputed as the new loan minus the dollars cashed out (the Freddie average amount cashed out and average balance imply a cashed-out share of a cash-out loan's new balance averaging $q_t \approx 0.23$). This gives about \$3.3 trillion. Excluding cash-out entirely is a conservative floor of \$2.1 trillion; also crediting the cash-out proceeds is a \$3.7 trillion ceiling; the central estimate is stable across $q_t \in [0.15, 0.35]$, within a few tenths of a trillion of the \$3.3 trillion headline. The one approximation is that cash-out borrowers, who may refinance when less in the money on rate, are assigned the rate-and-term refinancers' per-dollar gain, so the central estimate leans toward the upper part of the firm \$2.1 trillion floor.

Robustness. A deliberately conservative representative-model version of the per-dollar gain, varied over the discount, horizon, cost, and loan-age grid, brackets the cumulative between about \$0.9 trillion and \$3.0 trillion—large under every assumption (Appendix A.12, Appendix Table 4).

4.3 Ex ante value versus realized payoff: the surprise

The realized payoff so dwarfs the ex-ante value because the post-1990 rate path delivered far more than origination-date forward curves priced in—and, as the decomposition below shows, more than survey-consistent expectations predicted. Comparing, cohort by cohort over 1990–2016 origination cohorts with full 10-year realized windows ($N = 314$), the option value at origination with the value realized under optimal exercise on the actual PMMS path,

$$V^{\text{realized}} = V^{\text{ex ante}} + \text{Surprise},$$

the ex-ante value averages about 44 basis points while the realized optimal value averages about 89 basis points, a windfall of roughly 45 basis points (Figure 4). Of that gap, only about 3 bps reflects the term-premium wedge between forward rates and survey-expected rates—the component a forecast-consistent borrower would have anticipated—and a random-walk benchmark bounds the predictable part at 16 bps; the remaining 42 bps is residual path luck (Appendix N, Appendix Figure 12). The option was valuable at origination, but the secular decline in rates handed callable-debt households a gain far larger than the price—this is why the realized side, not just the ex-ante one, is historically important, and why the premium nets out roughly one-fifth to one-third of the gross payoff depending on the cost basis (the cumulative net Haig–Simons gain is about \$2.6 trillion netting the model-optimal premium, \$2.1 trillion netting the market-priced option cost, and \$1.7 trillion on the rate-and-term floor; Appendix A, Appendix Table 1).

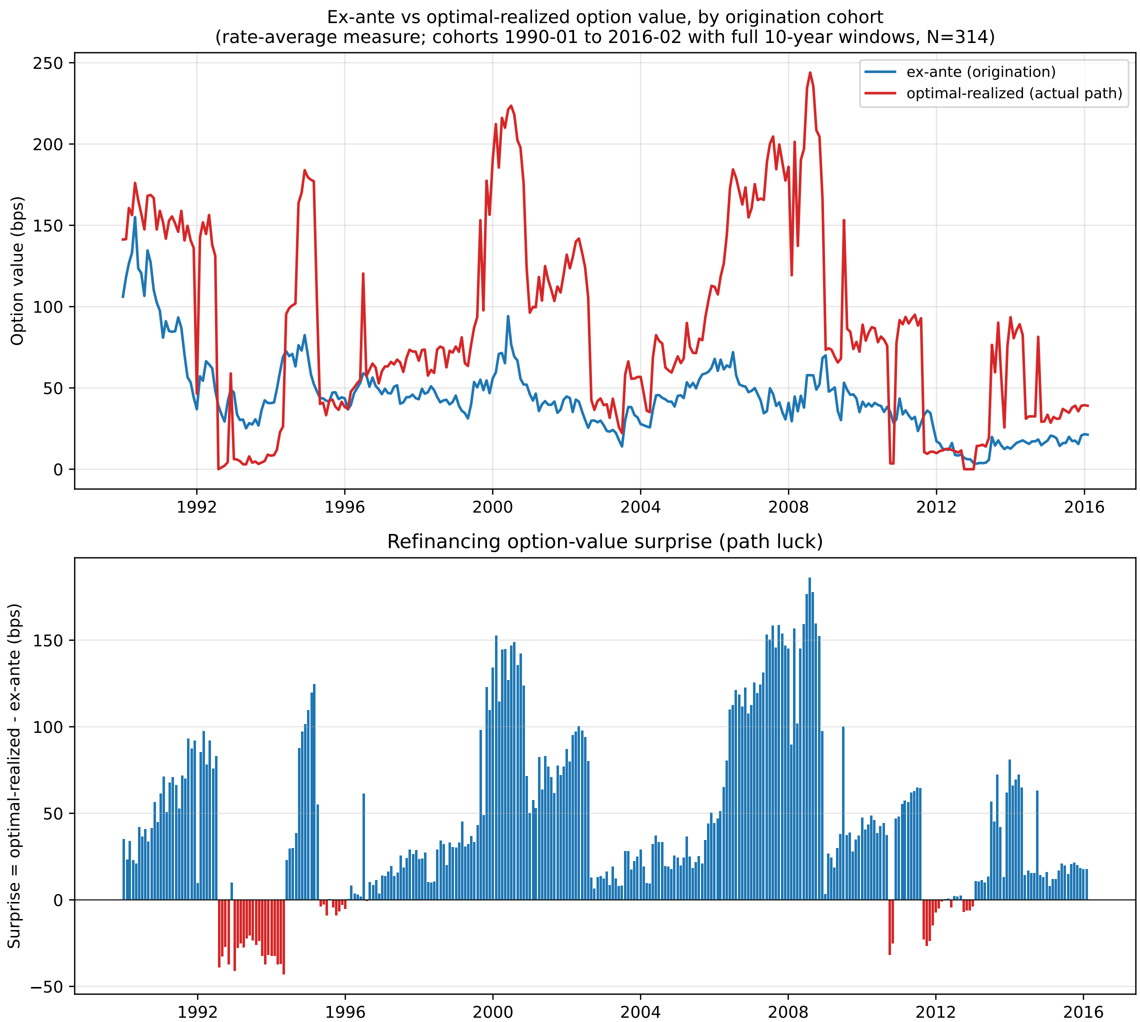

Figure 4: Ex-ante option value at origination versus the value realized under optimal exercise on the actual rate path, by cohort; the gap is the path-luck surprise. 1990–2016 origination cohorts with full 10-year realized windows ($N = 314$); rate-average measure on both series (the same cohort file underlies Appendix N). Reuses the paper's threshold and Monte-Carlo machinery; no new data.

4.4 Companion: the unrealized lock-in stock

The realized flow has a stock counterpart. When rates rise after a low-rate period, borrowers hold below-market liabilities they will not refinance; the gap between the unpaid balance and the market value of the liability, $B - L^{\text{callable}}$, is unrealized "lock-in" wealth, again absent from book net worth. Marking SCF mortgages to market, this stock is near zero through the falling-rate 1990s and 2000s but reaches about \$2.2 trillion in 2022 (about 17% of outstanding mortgage debt; 93% of mortgages below market). It is a distinct quantity from the realized flow (Section 2).

4.5 A transfer, not new wealth

The realized gain to borrowers is the mirror image of prepayment losses and lower reinvestment returns borne by the holders of the prepaid mortgages: holders of agency mortgage-backed securities, whole-loan investors, mortgage servicers, and portfolios with mortgage exposure, including the Federal Reserve through its holdings. This is the borrower-side image of the prepayment risk and negative convexity priced in the secondary market (Section 3.3, Appendix O). I therefore interpret G_t as a redistribution from mortgage investors to households with callable fixed-rate debt, not as an addition to aggregate wealth. The measurement point is narrower than "missing from the data" in general: the exercise payoff is not recorded as a realized capital gain or a liability revaluation in standard income or wealth statistics, even though it is a first-order, highly unequal transfer of economic resources.

5 Who Captured the Gains

This section measures who captured the realized gains—the money actually taken, the complement of what the behavioral literature documents as money left on the table: refinances that should happen but do not (Keys et al., 2016; Deng and Quigley, 2001; Andersen et al., 2020), the frictions that block them (DeFusco and Mondragon, 2020), and the groups whose exercise lags (Gerardi et al., 2023). Because refinancing depends on income, credit access, balance size, and attention, the realized gains of Section 4 accrued unevenly, and the shortfalls line up with the frictions that literature documents. I measure the distribution of the realized flow primarily from HMDA—the universe of mortgage originations, millions of loans per year—and use the GSE loan-level panel for the take-up frictions and the Survey of Consumer Finances for the wealth-rank and balance-sheet interpretation.

5.1 The distribution of realized gains

I allocate the aggregate gain G_t across groups by each group's share of that year's refinance origination volume, $G_{g,t} = G_t \times (\text{RefiVol}_{g,t}/\text{RefiVol}_t)$, using actual HMDA loan-level flows for every year 1990–2024 (the data described in Section 2.6). This volume-share allocation assigns each group the same per-dollar gain, so a group's share of the realized total equals its share of refinance volume—the dollar-weighted analogue of the take-up gaps below. The distribution is sharply regressive and uneven (Figure 5). By applicant income, the top quintile captured about 37% of the realized gain and the bottom quintile about 9%—a four-to-one gradient. By race, white borrowers captured about 78% of the gain, Asian and other borrowers about 10%, Hispanic borrowers about 7%, and Black borrowers about 4.4%, below their share of mortgage debt and population—the racial refinancing gap of Gerardi et al. (2023), here measured directly from realized dollar flows. The gains are also geographically concentrated: California alone accounts for about 25% of the national total. For 2018 onward, where HMDA reports the contract rate, I can replace the common per-dollar gain with each group's actual new rate; the resulting per-dollar gains are nearly identical across groups (about 5.8 versus 6.1 cents per dollar for white and Black refinancers in 2020, on the purpose-31 rate-and-term sample), because all groups refinance into the same market rates in a given year. The volume-share allocation is therefore a good approximation, and the ordering is unchanged. The allocation would have to be badly wrong to matter: given a gain-to-debt ratio of 0.59 (Table 4 below), Black refinancers' per-dollar gains would have to run roughly 70% above white refinancers' for Black capture to reach parity with the group's debt share, where the loan-level rates put the two within a few percent of each other.

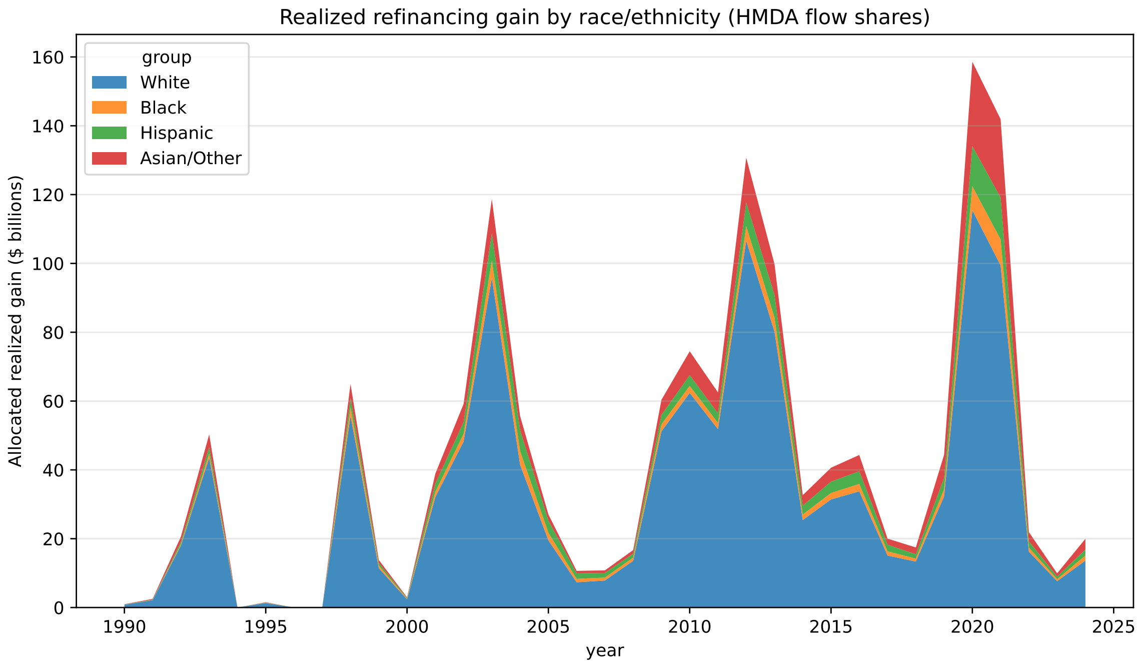

Figure 5: Realized refinancing gain by race/ethnicity, allocated from HMDA refinance volume shares, 1990–2024 (100% from actual HMDA loan-level data). Appendix Figure 4 shows each group's share of refinance volume over time.

All shares here are dollar-volume weighted, and the allocation uses rate-and-term dollar shares wherever HMDA distinguishes loan purpose (2018 onward). Reallocating by total-refinance shares moves no group's cumulative share by more than 1.6 percentage points, and because Black and Hispanic refinance dollars tilt toward cash-out, the disparities reported above are, if anything, slightly understated (Appendix A.12, Appendix Table 5).

5.2 Capture per dollar of mortgage debt

A share of gains is not by itself evidence of unequal exercise, because groups differ in how much mortgage debt they hold. Table 4 compares each group's gain share with its share of mortgage debt and of mortgage-holding households. The gain-to-debt ratio—each group's share of the realized gain divided by its share of mortgage debt, equivalently a dollar-weighted refinancing-flow-to-debt ratio since gains are allocated by volume—strips out unequal ownership: white borrowers are near parity (1.01) while the Black ratio is only 0.59, so per dollar of mortgage debt Black borrowers' refinancing flow was about two-fifths below parity; Hispanic borrowers are slightly below parity (0.94), and Asian and other borrowers above it (1.41), consistent with refinance volume concentrated in higher-balance markets. By income the bottom quintile's ratio is 0.82 against 1.11 for the fourth quintile, so the regressivity reflects both who holds mortgages and, conditional on holding one, who exercises the option. In dollar terms, had Black borrowers captured the realized gain in proportion to their 7% share of mortgage debt rather than their 4.4% share of the gain, they would have received roughly $86 billion more of the gross payoff over 1990–2024. The ratios are robust to building the denominator year by year—gain-weighted annual mortgage-debt shares interpolated across SCF waves, rather than the pooled share—which leaves the Black ratio at 0.60 (against 0.59 pooled) and the ordering unchanged.

Two patterns in Table 4 warrant comment. First, Asian and other borrowers capture above their debt share (1.41). Part of this is documented composition: their refinance volume tilts least toward cash-out (about 27% of refinance dollars, versus 37–40% for other groups; Appendix Table 5), so more of it counts toward the rate-and-term gain, and their mortgage debt is concentrated in recent, high-balance coastal-market cohorts—originated at rates that made the 2019–21 wave especially exercisable. Their per-dollar gains when refinancing are no larger than other groups' (shown above), so the ratio reflects exercise and composition, not pricing.

The second pattern is that the income gradient peaks at the fourth quintile (1.11) rather than the top (0.99); because top-quintile debt tilts toward jumbo and adjustable-rate products whose refinances fall partly outside the conventional rate-and-term universe measured here, the top-quintile ratio is sensitive to that selection and is best read as descriptive. Neither pattern is evidence on the Black–white gap, which Gerardi et al. (2023) attribute to exercise differences conditional on being in the money—consistent with the take-up and denial evidence below.

Table 4: Who captured the realized gain, relative to mortgage-debt denominators.

	Gain share	Mortgage-debt share	Holder share	Gain-to-debt ratio
By race/ethnicity				
White	78.4%	78.0%	78.5%	1.01
Asian/other	10.4%	7.4%	4.4%	1.41
Hispanic	6.8%	7.2%	7.2%	0.94
Black	4.4%	7.4%	9.9%	0.59
By income quintile				
Q1 (lowest)	8.6%	10.5%	19.5%	0.82
Q2	12.9%	13.6%	19.9%	0.95
Q3	17.4%	16.9%	19.9%	1.03
Q4	24.0%	21.6%	20.4%	1.11
Q5 (highest)	37.1%	37.3%	20.4%	0.99

Notes: Gain shares are allocated from HMDA refinance-origination dollar volumes, 1990–2024. Mortgage-debt and holder (mortgage-holding household) shares are from the Survey of Consumer Finances, pooled over the 1989–2022 waves (population weights, five implicates); race uses the wave-correct public-file coding (x6809 for 1998+, x5909 for 1989–95), with Asian and other/multiple-race responses reported together. Per-wave shares are benchmarked against Census/AHS homeownership-by-race envelopes in scf_share_diagnostics.csv. Holder share = share of mortgage-holding households (population-weighted count, not dollar-weighted). The gain-to-debt ratio is each group's gain share divided by its mortgage-debt share; below one means a group captured less than its proportional share of the gains. Generated by 20_denominators.py (config: scf_capture).

5.3 Take-up and access

Why the distribution is uneven is first a question about take-up—the probability an in-the-money borrower actually exercises. I estimate this from a discrete-time hazard model on a Freddie Mac loan-month panel (about three million sampled loan-months), regressing voluntary prepayment on the refinancing incentive (a spline), FICO, loan-to-value, loan balance, loan age, burnout, and calendar and geographic controls, and attributing the in-the-money excess over the deep-out-of-the-money baseline to refinancing (Appendix Figure 2).

Take-up responds strongly to the incentive (the estimated S-curve replaces the paper's earlier hard threshold), and the friction gradients are economically large: over a twelve-month in-the-money window, cumulative take-up rises with credit score (from about 31% at FICO below 660 to 37% at 740–780) and falls steeply with loan balance (from about 51% at a $29,000 balance to 29% at $373,000), with a seasoning hump around three years. The GSE samples carry no borrower race or income, so the racial and income take-up gaps remain sourced from HMDA and the SCF; this model supplies the friction structure through which those gaps operate.

Take-up measures whether an in-the-money borrower chooses to refinance; a distinct friction is whether a borrower who applies is approved. HMDA records the disposition of every application. Pooling the roughly 185 million refinance applications acted upon over 1990–2024, Black applicants were denied at 41.1% and Hispanic applicants at 33.1%, against 20.5% for white applicants (Table 5; Appendix Figure 3 traces the gap over time)—a roughly two-to-one gap in denial rates that compounds the take-up gap above and persists in the modern reporting regime (32.6% versus 16.0% over 2018–2024, reported separately because HMDA's universe and race coding changed in 2018). These are unadjusted rates, not estimates of disparate treatment: applicants differ in credit characteristics, income, loan size, and location, and Bhutta et al. (2025) find that observable risk factors account for most of the raw denial gap in adjusted comparisons. For the incidence question here, however, the unadjusted rate is the relevant quantity—whatever its source, a denied application is a gain not captured. The realized-gain shortfall for minority borrowers therefore reflects both ends of the pipeline: they apply to refinance less often when in the money, and are turned down more often when they do.

Table 5: Refinance denial rates by race, 1990–2024 (HMDA).

	Applications (mn)	1990–2024	1990–2017	2018–2024
White	142.8	20.5%	21.2%	16.0%
Black	14.1	41.1%	42.5%	32.6%
Hispanic	15.9	33.1%	34.6%	25.7%
Asian/other	11.8	24.2%	26.2%	16.9%

Notes: Refinance applications acted upon—originated or denied—by race, from the HMDA Loan/Application Register: full-record local archives for 1990–2017 (loan purpose = refinance) and the CFPB data-browser API for 2018–2024 (purposes 31+32, matching the legacy definition; purpose-31-only rates are within 1 pp). The 2018–24 universe excludes open-end lines of credit and reverse mortgages, which the pre-2018 LAR did not report; race/ethnicity uses the same harmonized mapping as the volume-share analysis (Hispanic ethnicity takes precedence). Denial rate = denied / (originated + denied); race-not-reported applications excluded. The 2018–24 sub-period is reported separately because the reporting regime changed in 2018. Generated by 25_denials_combined.py.

The SCF supplies what HMDA cannot: the position of refinancing households in the wealth distribution, their birth cohorts, and the book-versus-market balance-sheet interpretation. Allocating the realized flow across SCF households (recent refinancers, population-weighted over the five imputation implicates) reproduces the regressive income pattern and the below-parity gain-to-debt ratios of Black and Hispanic households—an SCF-only Black ratio of 0.66, averaged across waves, independently corroborating the HMDA-based 0.59. The SCF cross-sectional "take-up" measure—of each group's in-the-money balance, the fraction that refinanced—is retained as a robustness exhibit; the GSE loan-month hazard is the primary take-up measure.

What the estimates can and cannot identify. Three measurement limits bound the estimates of this section and the last—the modeled pre-1999 per-dollar gain, the GSE panel's lack of borrower demographics, and the SCF's triennial frequency—detailed in Appendix A.10. The distributional ordering is robust to all three; the dollar magnitudes should be read as estimates, bracketed by the ranges in Appendix Table 4.

6 Conclusion

This paper measures both sides of the refinancing option from the borrower's vantage. Ex ante, a borrower-side option-adjusted cost rate—built from the optimal refinancing rule, a calibrated term-structure model, and Monte Carlo simulation over 434 origination months—implies that the option lowers expected financing cost by about 31 basis points on average, rising to 110 bps in the 2022–2024 rate spike and 143 bps in late 2023, with rate volatility the strongest driver; the series co-moves with the prepayment component priced in mortgage-backed-securities spreads. Ex post, treating the mortgage as a callable liability, households realized roughly \$3.3 trillion in present-value capital gains, in constant 2026 dollars, over 1990–2024 (about \$2.1–\$2.6 trillion net of the ex-ante option value depending on the cost basis; \$2.1 trillion excluding cash-out refinances). That payoff was far more than origination-date prices anticipated, and for most cohorts more than even survey-consistent rate expectations would have predicted (Appendix N). It was matched one-for-one by investors' prepayment losses, with the dollars concentrated among higher-income households through their larger mortgage holdings, and Black borrowers capturing well below a proportional share even per dollar of mortgage debt. A companion \$2.2 trillion stock of locked-in below-market mortgages in 2022 is a current mark, not a realized flow.

Three implications follow. First, for borrowers, their advisors, and affordability debates: the contract rate determines the monthly payment and underwriting, but it is not the loan's expected long-run cost. When rates rose from about 3% to 6.6% during 2022–2023, the deterioration in payment-based affordability was real and immediate, but headline-rate comparisons overstate the rise in expected long-run financing cost, because they ignore the option that becomes most valuable precisely when rates are high. A 6.6% borrower does not have a cheap mortgage; she has an expensive payment and a valuable embedded option. Existing consumer tools omit this distinction: rent-versus-buy calculators, payment calculators, and affordability estimators treat the rate as fixed for thirty years,[2] and refinance

[2]The *New York Times* Upshot calculator applies a single user-entered mortgage rate to the entire analysis horizon. Zillow's methodology document assumes "the chosen constant interest rate and loan duration of 30 years." Bankrate acknowledges the omission explicitly: "the ability to refinance a mortgage for a lower rate

calculators are separate and reactive—they ask whether to refinance now, not what the mortgage is expected to cost at origination. Making the option-adjusted cost rate visible—alongside the contract rate and the annual percentage rate—would sharpen both individual mortgage decisions and the public understanding of housing affordability, a gap that matters most in a market where many borrowers already struggle to shop for the rate itself (Bhutta et al., 2026). Second, comparisons between fixed- and adjustable-rate mortgages based on initial rates miss the embedded asymmetry—the fixed-rate borrower is protected if rates rise and can still benefit, by refinancing, if they fall—though a full comparison would require modeling ARM caps, margins, and reset risk under the same rate process (Campbell and Cocco, 2003; Koijen et al., 2009). Third, the realized-gain accounting gives the refinancing channel of monetary policy transmission (Berger et al., 2021; Eichenbaum et al., 2022) a cumulative dollar magnitude: the transfer from mortgage investors to households, and its uneven distribution, is the incidence of that channel over three decades of falling rates.

Two caveats bound these estimates. The option values assume optimal exercise and are therefore an upper bound, since real borrowers face inattention, inertia, and credit constraints (Keys et al., 2016; Gerardi et al., 2023); and the largest modeling sensitivity, the mortgage–Treasury spread specification, shifts average option values by up to 17 basis points while preserving the time-series pattern. The realized-gain magnitudes rest on modeled pre-1999 per-dollar gains and survey-based allocations, so they are best read as estimates, with the ranges reported in the appendix.

Even so, the wedge between the contract rate and the expected financing cost is economically meaningful, state-dependent, and consistent with how investors price prepayment risk; and the realized-gain accounting shows that the exercise of the refinancing option has been a first-order, highly unequal transfer that standard income and wealth statistics miss.

. . . was not factored in." The same is true of calculators from Fidelity, SmartAsset, Schwab, and AARP.

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A Data and Methodology for Realized Refinancing Capital Gains

This appendix documents the construction of the realized refinancing capital-gain series of Sections 4 and 5: the quantities estimated, the per-dollar gain formula, every data source and its coverage, the assumptions and imputations (with the period each applies to), the cross-checks used to validate the series, the limitations, and the code that reproduces it. The guiding principle is that the headline magnitudes for 1994–2024 rest on observed data, and every place where a value is modeled, held, or proxied is stated explicitly.

A.1 Quantities estimated

I construct four related quantities and keep them conceptually separate.

1. **The realized capital-gain flow G_t** (headline): the present-value gain captured by borrowers who refinanced an existing loan in year t—rate-and-term loans in full plus the old-balance portion of cash-out refinances—aggregated to a national annual series, 1990–2026.

2. **The ex-ante/ex-post "surprise"**: the difference between the option value priced at origination and the value realized along the actual rate path, which isolates how much of the realized gain was unanticipated.

3. **The distribution of G_t across households**: who *earned* the realized gain—by income, race, and geography from HMDA loan-level flows (1990–2024); by wealth rank and birth cohort from the Survey of Consumer Finances (SCF); with take-up frictions estimated from the GSE loan-month panel.

4. **The unrealized lock-in stock** (companion): the point-in-time gap between the unpaid balance and the market value of outstanding mortgage liabilities, $B - L^{\text{callable}}$, from the SCF.

The flow (1) and the stock (4) are distinct quantities—a cumulated payoff versus a current mark—and are never added together.

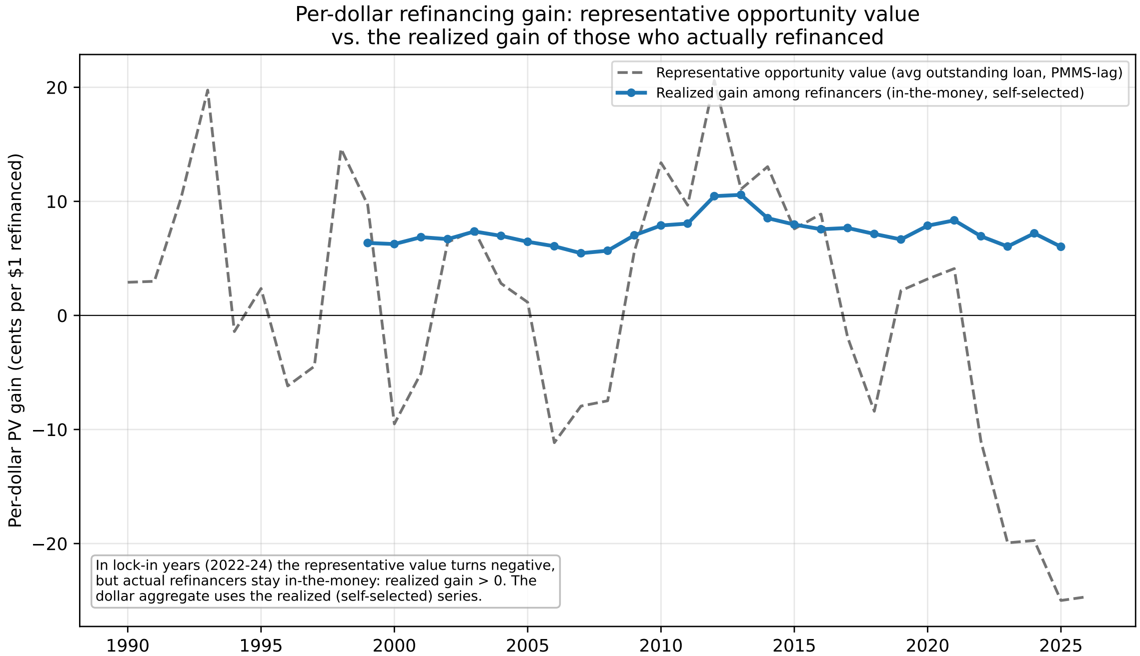
A.2 The per-dollar gain

For a refinance from old coupon c_{old} to new coupon c_{new} on balance B with remaining term n_{old} months, refinanced into a fresh $n_{\text{new}} = 360$-month loan at cost $K = \text{fixed} + \text{points} \times B$, the headline per-dollar gain is the *terminal-balance-adjusted* present value

$$g^{\text{TBA}} = \frac{1}{B}\Big[\underbrace{\textstyle\sum_m P^{\text{old}}\, d^m + B_H^{\text{old}}\, d^H}_{\text{PV of keeping the old loan to } H} - \underbrace{\big(\textstyle\sum_m P^{\text{new}}\, d^m + K + B_H^{\text{new}}\, d^H\big)}_{\text{PV of the new loan}} \Big], \qquad (2)$$

where $P^{\cdot}$ is the scheduled monthly payment, $B_H^{\cdot}$ the balance remaining at horizon H, and $d^m = e^{-\rho m/12}$ the (continuous) discount factor, with ρ fixed at 5% in the baseline and varied in the scenario fan below. Including both terminal balances charges the borrower for the higher residual principal of the re-amortized new loan ($B_H^{\text{new}} > B_H^{\text{old}}$), which neutralizes the maturity-extension that inflates a raw payment-savings measure.

I report two alternatives for robustness: an option-theoretic measure $g^{\text{OT}} = [\text{MV}_{\text{old}} - (B + K)]/B$, where MV_{old} is the market value of the old loan's remaining cash flows discounted at the prevailing market rate (this needs no ρ or H); and a naive annuity measure g^{simple} omitting the terminal-balance term, included only to quantify the maturity-extension bias. Baseline parameters: $\rho = 5\%$, $H = 120$ months, costs = 1 point + \$2,000, representative $B = \$250{,}000$. A scenario fan varies $\rho \in \{3, 5, 7\}\%$, $H \in \{60, 120, 360\}$, the loan age, and costs.



Appendix Figure 1: Two distinct per-dollar series. The *representative opportunity value* marks the average outstanding loan to market and turns negative in lock-in years (2022–24) when the typical borrower is out of the money; the *realized gain among refinancers* (GSE loan-level, 1999+) stays positive throughout because households that refinance are positively selected into being in the money. The aggregate flow G_t uses the latter.

The aggregate flow, on the headline existing-loan basis, is

$$G_t = \max(g_t, 0) \times V_t^{\text{exist}}, \qquad V_t^{\text{exist}} = V_t^{\text{rt}} + \big(1 - q_t\big)\, V_t^{\text{co}}, \tag{3}$$

where V_t^{tot} is total refinance origination volume; s_t^{co} is the cash-out *incidence* (its share of that volume), so $V_t^{\text{rt}} = (1 - s_t^{\text{co}})V_t^{\text{tot}}$ is rate-and-term volume and $V_t^{\text{co}} = s_t^{\text{co}} V_t^{\text{tot}}$ is cash-out volume; and q_t is the cashed-out share of a cash-out loan's new balance, so $(1 - q_t)V_t^{\text{co}}$ is the old-balance portion of cash-out refinances. The rate-and-term floor $\max(g_t, 0)V_t^{\text{rt}}$ excludes cash-out entirely and the ceiling $\max(g_t, 0)V_t^{\text{tot}}$ also credits the cash-out proceeds. Every dollar figure is expressed in constant 2026 dollars: each year's G_t, a present value in that refinance year's dollars, is scaled by the ratio of the 2026 to the year-t PCE price index before cumulating, so the 1990–2024 total is a single constant-dollar magnitude. The floor $\max(g_t, 0)$ encodes self-selection: rate-and-term refinancing essentially vanishes when it is not in the money, so years with a negative model gain (rising rates, e.g. 2022–2024) contribute approximately zero realized gain rather than a negative number.

Net of the premium (the headline capital gain). Equation (3) is the *gross* realized payoff. Because a callable mortgage carries a higher contract rate than an otherwise-identical non-callable loan, the borrower pays a premium for the refinancing option, and the economically correct capital gain is payoff minus this cost basis. The premium is the paper's ex-ante option value; I convert it to a per-dollar present value with the same machinery (the per-dollar PV of a rate reduction equal to the ex-ante option value, on a fresh 30-year loan at origination), by origination year, and match it to each refinance year through the loan's median age:

$$g_t^{\text{net}} = g_t - p_{\,t-a_t}, \qquad G_t^{\text{net}} = \mathbf{1}\{g_t > 0\}\, g_t^{\text{net}}\, V_t^{\text{rt}}, \tag{4}$$

where p_y is the per-dollar premium for origination year y and a_t is the median age of loans refinanced in t. I report G^{net} as the headline Haig–Simons capital gain and the gross G as an upper bound. Empirically the premium nets out about one-fifth of the gross (the cumulative falls from roughly \$3.3 to \$2.6 trillion), because the realized post-1990 rate decline far exceeded what was priced ex ante — most of the realized gain is an unanticipated windfall. This makes the section's three numbers one consistent quantity: the ex-ante option value is the cost basis, the realized PV reduction is the payoff, and their difference is the capital gain.

Alternative cost basis: the market price of the option. The model premium is the *optimal-exercise* value implied by the forward curve. Borrowers, however, paid the *market* price of the option in their contract rate. Appendix Table 1 reports the net gain when the basis is instead the market option cost—the OAS-netted prepayment component of the secondary spread (Appendix O)—converted to a per-dollar present value with the same machinery as p_y. The market basis *exceeds* the model-optimal basis in most years with OAS data: the market price embeds a prepayment-risk premium on top of the forward-curve-implied option value. Netting the market basis therefore lowers the cumulative net gain from \$2.6 trillion to about \$2.1 trillion (ratio backfill before 1997; \$2.4 trillion under a zero pre-1997 basis), so the headline net figure is the *upper* end of the basis range and the gross payoff is unaffected. Figure 3 shades the band between the two bases.

Appendix Table 1: Net realized gain under alternative option-premium cost bases, 1990–2024 (constant 2026 dollars).

Cost basis	Net gain (existing-loan, \$tn)	Net rate-term floor (\$tn)
Model-optimal premium (headline)	2.6	1.7
Market option cost, ratio backfill pre-1997	2.1	–
Market option cost, zero basis pre-1997	2.4	–

Notes: The model-optimal basis nets out the paper's ex-ante option value (forward-curve-implied, optimal exercise). The market basis nets out the OAS-netted prepayment component of the secondary spread (current-coupon yield minus the 10-year Treasury, minus the option-adjusted spread; Bloomberg, monthly from September 1996), converted to a per-dollar present value with the same machinery as the model premium; negative months floored at zero. Pre-1997 months have no OAS data and are bracketed by two rules: the 1997–99 average market-to-model ratio applied to the model premium, and a zero basis. The market basis exceeds the model basis in most years—the market price embeds a prepayment-risk premium—so the model-optimal basis yields the larger net gain. Generated by 08_realized_gains_figures.py.

A.3 Data sources

Appendix Table 2 lists each data source, its coverage, and its role in the analysis.

Appendix Table 2: Data sources. All are public; the GSE loan-level data require a free click-through license. MBA estimates are themselves model-based but are the standard national series and are benchmarked to HMDA.

Series	Source	Coverage	Role
Refinance $ volume	MBA Historical Origination Estimates	1990–2026 (annual)	Headline natio tor
validation	HMDA loan-level (LAR; CFPB historic; API)	1990–2024	Volume cross-ch
Distribution of flows	HMDA loan-level (race/income/geography)	1990–2024	Who earned t mary)
Take-up frictions	Freddie loan-month panel (hazard)	1999–2024	Exercise pro FICO/LTV/bal
Cash-out share	Freddie Quarterly Refinance Report (Total US)	1994–2017 (annual)	Rate-term/cash
modern split	HMDA (loan-purpose 31 vs 32)	2018–2024	Split (purpose f
Per-dollar gain	Freddie Single-Family Loan-Level	1999–2024	Actual rates/balances/
Median loan age	Freddie Quarterly Refinance Report	1994–2016	Old-rate lag in t
Mortgage rate (PMMS)	Freddie PMMS via FRED (`MORTGAGE30US`)	1971–	Old/new rate p
Household detail	SCF full public microdata	1989–2022 (triennial)	Stock + who-ea bution
Mortgage debt outstanding	FRED `HHMSDODNS`	quarterly	Scaling / SCF c
MBS spread anchor	NY Fed refinance-boom estimates	2020–21	External magni

Volume. The national denominator is the MBA refinance-origination series, used for every year so the aggregate has no source discontinuity. As an independent check I sum originated refinance loans (loan purpose = refinancing, action = originated) directly from the HMDA Loan/Application Register files; the HMDA total tracks MBA within roughly 10–20% in every overlapping year.

Cash-out split. The headline existing-loan basis credits a cash-out refinance only for its existing-loan component: the rate reduction on the old balance is a genuine refinancing gain, while the cash-out proceeds are home-equity extraction—new borrowing at the new rate—and earn no gain (eq. (3)); the rate-and-term floor excludes cash-out loans entirely. The split comes from HMDA's loan-purpose field for 2018–2024 (which distinguishes refinancing from cash-out refinancing) and from the Freddie Quarterly Refinance Report's annual "Total US" cash-out incidence for 1994–2017. Freddie defines cash-out as a new loan balance at least 5% above the refinanced loan's unpaid balance, on conventional prime matched refinances; this differs from HMDA's purpose-field definition, creating a modest definitional seam at 2018 that I flag and treat as a robustness margin.

Per-dollar gain. For 1999–2024 the gain in (2) uses each loan's *actual* note rate, balance, and remaining term from the Freddie Single-Family Loan-Level data; the new rate is the prevailing PMMS at the payoff month (a market proxy, validated against the realized new rate for the HARP cohort). For 1990–1998, where no public loan-level data exist, the gain is a model value: the old rate is PMMS lagged by the median loan age and the new rate is the contemporaneous PMMS.

A.4 Layer 1: the aggregate flow

The per-dollar gain g_t is the balance-weighted actual gain from the loan-level data where available (1999–2024) and the model gain otherwise (1990–1998). It is multiplied by rate-and-term volume per (3), and cumulated. A scenario fan brackets the series under the parameter grid above.

A.5 Layer 2: loan-level micro (GSE)

From the Freddie Single-Family Loan-Level sample I identify voluntary payoffs (zero-balance code 01) on fixed-rate loans. Because the data do not label a payoff as a refinance or a home sale, I attribute payoffs to refinancing by a rate-gap rule: a payoff is a refinance if the loan was in the money by at least 75 basis points (note rate minus prevailing PMMS) at payoff; home sales occur roughly independently of the rate gap. For attributed refinances I compute

(2) at the payoff date using the loan's actual balance and remaining term, and report the balance-weighted per-dollar gain by payoff year and by FICO and LTV bucket. I report sensitivity of the attributed share to the threshold (0, 50, 75, 100 bps).

Take-up hazard. To replace the hard rate-gap threshold with an estimated take-up elasticity, I build a discrete-time hazard panel from the same loan-level samples: positive-balance loan-months are the at-risk spell-months and the event is a voluntary payoff (zero-balance code 01), attached to the *last* positive- balance month so its covariates are not read off the zero-balance record. I case-control sample the at-risk months (all events, all in-the-money months, and a weighted subsample of the rest) and fit a logit of the event on a spline in the refinancing incentive, FICO, LTV, the CLTV gap, log balance, a loan-age spline, a burnout proxy, and calendar-year and state fixed effects. The refi-attributed hazard is the fitted hazard net of the deep-out-of-the-money (move) baseline, and twelve- month take-up is the cumulative refi-attributed exercise probability, tabulated by FICO, LTV, balance decile, loan age, and state. Estimated take-up rises with FICO and falls steeply with loan balance, and the incentive enters as an S-curve; because the public GSE samples carry no race or income, this layer supplies the friction structure, not the racial/income gap.

A.6 Layer 3: ex-ante vs ex-post surprise

For each origination cohort I compare the option value priced at origination (the model's ex-ante value) with the value realized under optimal exercise along the actual PMMS path (the backtest). The gap is the "path-luck" surprise. This layer reuses the paper's existing threshold and Monte-Carlo machinery and adds no new data.

A.7 Layer 4: who earned the gain, and the lock-in stock (SCF)

Mark-to-market. For every SCF mortgage I value the liability as $L^{\text{callable}} = \min\{\text{PV(remaining scheduled pa}$ $K\}$, using the household's reported coupon, balance, and remaining term, and define hidden wealth $B - L^{\text{callable}}$. The min is the refinancing option: a borrower never lets the liability's market value exceed the cost of refinancing it away. Aggregated by wave this is the lock-in stock; by age, race, and income quintile it is its distribution. I benchmark total SCF mortgage debt against FRED `HHMSDODNS` (coverage rises from 0.67 in 1989 to 0.91 in 2022).

Who earned the flow (HMDA, primary). The distribution of the realized flow by race, income, and geography is allocated from actual HMDA refinance-volume shares, $G_{g,t} = G_t \times (\text{RefiVol}_{g,t}/\text{RefiVol}_t)$, for every year 1990–2024 (no interpolation). Three coding eras are harmonized: the pre-2004 files (a single combined race field, code 4 = Hispanic); the 2004–2017 files (separate applicant ethnicity and race, code 4 = Native Hawaiian); and the 2018+ modern files (derived race/ethnicity, plus the contract rate, used to refine each group's new rate). Withheld-race volume is reported separately and excluded from the share denominator; income groups are per-year count-weighted quintiles. The pre-2007 files use deflate64 compression and are streamed with a pure-Python decoder. State FIPS codes are mapped to postal abbreviations so geography aligns across eras.

Who earned the flow (SCF, wealth rank and cross-check). I identify recent refinancers (the first-mortgage refinance flag, with the current loan obtained within three years of the survey), take their refinanced balance and characteristics, and apply the year's per-dollar gain to allocate the realized flow across income quintiles, race, and birth cohort—an independent corroboration of the HMDA flows and the source for the *wealth*-rank dimension HMDA lacks. SCF variables used: coupon (`X816`), balance (`X805`), term (`X806`/`X815`), year obtained (`X802`), refinance flag (`X7137`), age (`X14`), race (`X6809`), income (`X5729`), and the analysis weight (`X42001`). All statistics use the five imputation implicates and the survey weight.

A.8 Assumptions and imputations

Appendix Table 3 is the complete audit of what is observed versus modeled, by period. The headline 1994–2024 magnitudes rest on observed volume and observed cash-out split; the only modeled inputs touching that window are the new-rate proxy in the loan-level gain (HARP-validated) and the refinance/move attribution rule.

Appendix Table 3: Observed vs. modeled inputs by period. *No public, consistently defined cash-out split exists before 1994 (Freddie's matched-refinance series begins in 1994); for 1990–1993 the cash-out split is held at its 1994 value, so those years enter the headline as a modeled component rather than an independent measurement. †The actual new rate is observed only for the HARP cohort (2009–2016), against which the PMMS proxy is validated.

Input	1990–1993	1994–1998	1999–2024
Refi volume	MBA (HMDA-checked)	MBA (HMDA-checked)	MBA (HMDA-checked)
Cash-out share	held at 1994 *	Freddie (observed)	Freddie / HMDA (observed)
Old rate	PMMS lag (model)	PMMS lag (model)	loan-level (actual)
New rate	PMMS (model)	PMMS (model)	PMMS at payoff (proxy)†
Median loan age	held at 1994 (2.7y)	Freddie (observed)	loan-level (actual)
Refi vs. move	—	—	75 bp rate-gap rule

Additional assumptions, all reported in sensitivity: the discount rate ρ, the horizon H, the new-loan term (a fresh 30-year), and refinancing costs are modeling parameters spanned by the scenario fan; the representative balance affects only the per-dollar cost share, not the aggregate (which scales with observed dollar volume). The pre-1999 model gain is not free-form: fed the observed median loan age, it reproduces the loan-level estimate where the two overlap (e.g. 2003: 0.075 model vs. 0.074 loan-level), so the early-period series is calibrated to, not merely assumed away from, the micro data.

A.9 Validation and reconciliation

- **Volume:** HMDA-summed refinance originations match the MBA series within ∼10–20% annually, 1990–1999 and 2018–2024.
- **Per-dollar gain:** the model gain (with observed median age) and the loan-level actual gain agree where they overlap (2003: 0.075 vs. 0.074; 2020–2021: 0.064 model vs. 0.065–0.067 loan-level).

- **Cash-out split:** the Freddie annual series matches the Freddie matched-refinance research-note values (1998: 0.484 vs. 0.486; 2003: 0.363 vs. 0.366; 2006: 0.868 vs. 0.867).
- **External anchor:** for 2020–2021 the New York Fed estimates rate refinances cut annual housing costs by about $24 billion; capitalized over the horizon this is consistent with the ∼$300 billion present-value gain (in then-current dollars, to match the Fed's nominal flow; not deflated to 2026 dollars) I estimate for that episode (flow vs. present value).
- **SCF coverage:** total SCF mortgage debt is 0.67–0.91 of FRED `HHMSDODNS`, the expected survey-coverage range.
- **Self-selection:** in lock-in years where the representative model gain is near zero, the loan-level data confirm that the households who *did* refinance were in the money with positive gains.

A.10 Limitations

No public loan-level mortgage data exist before 1999, so the pre-1999 per-dollar gain is necessarily a (calibrated) model rather than a measurement. No public, consistently defined cash-out/rate-term split exists before 1994, so for 1990–1993 the split is held at its 1994 value—a modeled component of the headline (flagged in Appendix Table 3), not an independent measurement. The new rate in the loan-level gain is a market (PMMS) proxy except for the HARP cohort, because the public GSE data do not link a paid-off loan to its replacement. The refinance/move attribution is a rule, not an observed field. The Freddie loan-month panel carries no borrower race or income, so the distributional shares are sourced from HMDA volumes and the SCF rather than from the take-up model directly. MBA volume is itself a model-based estimate (benchmarked to HMDA). The SCF is triennial, so the who-earned-it allocation is built from recent refinancers in each wave rather than a continuous flow, and the cash-out definition differs between the Freddie (5%-balance) and

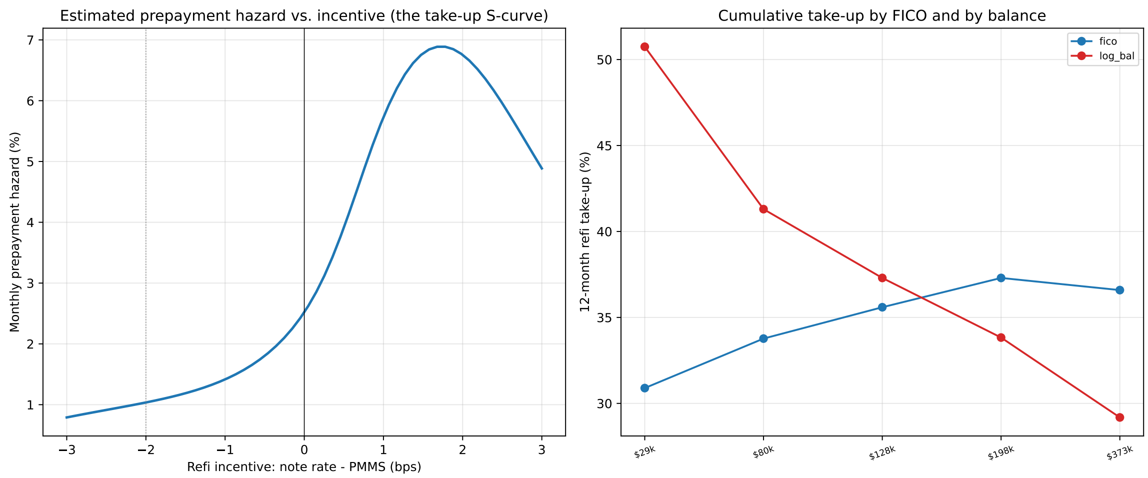
HMDA (loan-purpose) sources; the pre-2018 group allocation pools rate-and-term with cash-out volume, a bias bounded at roughly half a percentage point of any group's cumulative share (Appendix Table 5). Finally, the realized gain is a transfer from mortgage investors to borrowers, not an addition to aggregate wealth.

A.11 Reproducibility

The analysis is a self-contained pipeline that reuses the paper's valuation code and writes only its own outputs. Stages: fetch external data (FRED, HMDA API); parse the HMDA LAR files, the Freddie loan-level samples, the Freddie Quarterly Refinance Report, and the SCF; compute the per-dollar gain primitives; build the aggregate flow, the loan-level micro, the surprise, and the SCF stock and flow attribution; and assemble the figures and tables. Every figure and table is regenerated from the public inputs by a single orchestrator script, and each intermediate series is written to a documented CSV.

A.12 Take-Up Hazard and Robustness Exhibits

Appendix Figure 2 reports the estimated take-up hazard from the Freddie Mac loan-month panel, and Appendix Table 4 brackets the cumulative realized gain under the representative-model per-dollar measure of Section 4.

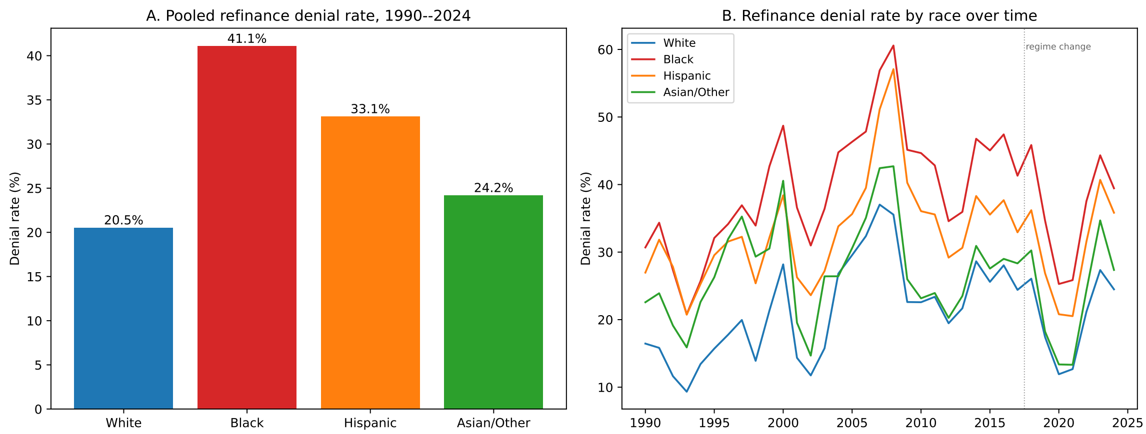


Appendix Figure 2: Estimated prepayment-hazard "take-up" S-curve in the refinancing incentive (left) and cumulative twelve-month take-up by FICO and by loan balance (right), from the Freddie Mac loan-month panel.

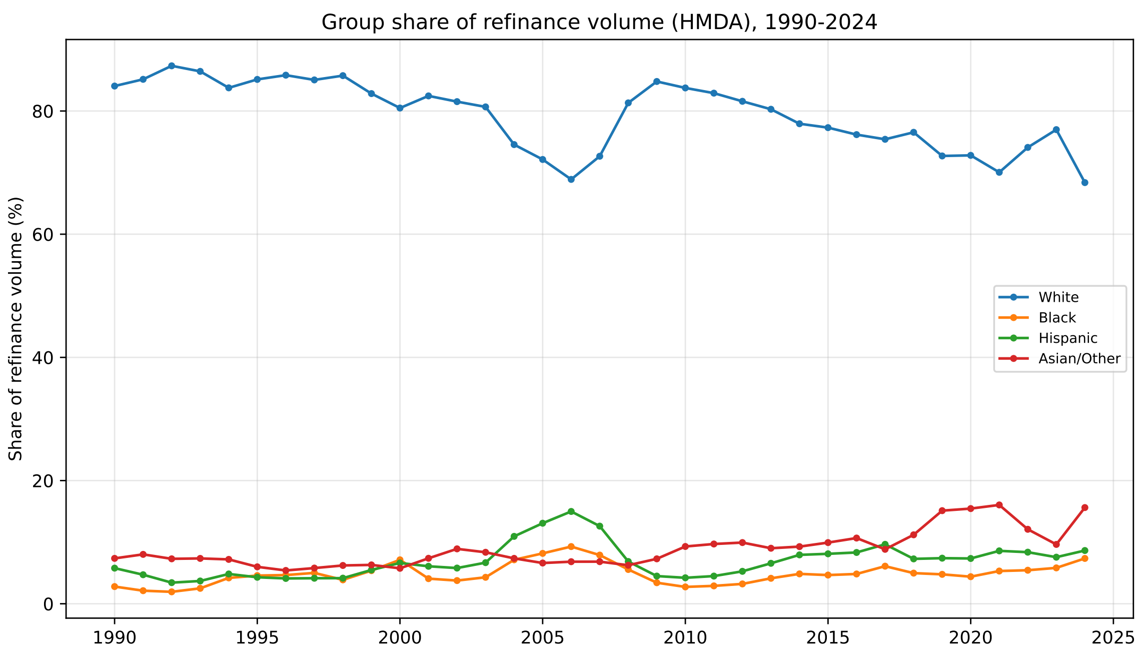
Appendix Table 4: Sensitivity of the cumulative realized refinancing capital gain under the conservative representative-model per-dollar measure, 1990–2024 ($ billions, constant 2026 dollars).

Variation	Cumulative gain ($bn)
Baseline (terminal-balance-adjusted gain, 5% discount)	1878
Market discount (prevailing mortgage rate)	1873
All refinances, rate effect only (incl. cash-out)	2830
Option-theoretic gain (market value minus par)	3022
Naive annuity (no terminal-balance adjustment)	2277
Discount rate 3%	2052
Discount rate 7%	1720
Horizon 5 years	934
Horizon full term (30 years)	3048
Closing costs 0.5% + $2,000	1992
Closing costs 1.5% + $3,000	1673
Median loan age 2.5 years	1361
Median loan age 5.0 years	1951

Notes: Each row is the cumulative realized gain $\sum_t G_t$, 1990–2024, under one variation of the per-dollar gain measure, discount rate, horizon, closing costs, or median loan age of Appendix A.2. The baseline is the terminal-balance-adjusted measure (eq. (2)); the option-theoretic (market value minus par) and naive-annuity (no terminal-balance adjustment) rows are the alternative per-dollar measures defined there. This representative-model series is deliberately conservative and sits below the headline loan-level series of Figure 3; see Section 4.2. Generated by 03_layer1_aggregate.py.



Appendix Figure 3: Refinance denial rates by race, 1990–2017 (HMDA full-record LAR). *Left:* pooled denial rate by race. *Right:* the denial rate by race over time. Denial rate = denied / (originated + denied) among refinance applications with race reported. The 2018–2024 modern-regime rates are reported separately in Table 5.



Appendix Figure 4: Each group's share of HMDA refinance dollar volume over time, 1990–2024; the companion to Figure 5's allocated-gain levels.

Cash-out composition detail. The headline aggregate is rate-and-term only, and for 2018 onward the group allocation already uses purpose-31 (rate-and-term) dollar shares;

before 2018 HMDA does not distinguish purpose within refinances, so shares come from all refinance volume. Appendix Table 5 compares the two bases on 2018–24 data: reallocating G_t by total-refinance shares moves no group's cumulative share by more than 1.6 percentage points. Black and Hispanic borrowers tilt toward cash-out (about 40% of their refinance dollars, versus 37% for white and 27% for Asian/other borrowers), so an all-refinance allocation slightly *overstates* their share of the rate-and-term gain. Applying the 2018–24 propensity gaps to the Freddie-era aggregate cash-out share (47% of 1994–2017 volume) bounds the pre-2018 overstatement of the Black and Hispanic allocated shares at roughly 10% of those shares ($\sim$0.5 percentage points of the total), so the main-text disparities are, if anything, slightly understated.

Appendix Table 5: Cash-out composition robustness: allocation of the 2018–24 rate-and-term gain under alternative share bases.

	Total-share alloc.	Rate-term alloc.	Δ share	Cash-out propensity
White	73.0%	72.0%	-1.01 pp	37%
Asian/other	13.5%	15.1%	+1.59 pp	27%
Hispanic	8.3%	7.9%	-0.40 pp	40%
Black	5.2%	5.0%	-0.19 pp	40%

Notes: HMDA 2018–24, dollar-volume weighted. "Total-share" allocates the annual rate-and-term gain G_t by each group's share of all refinance volume (purposes 31+32), the only basis available before 2018; "rate-term" uses purpose-31 volume only (the basis actually used for 2018+). Cash-out propensity is the purpose-32 dollar share of each group's total refinance volume. Freddie-era aggregate cash-out share (1994–2017 mean): 47%. Generated by 23_cashout_composition.py.

B Optimal Refinancing Threshold

The borrower holds a mortgage with current rate i and remaining balance M. Refinancing incurs a tax-adjusted cost κ. The mortgage rate follows a Brownian motion with volatility σ. The borrower has a real discount rate ρ (fixed at 0.05 in this appendix's replication; the headline series uses the observable ρ_t), faces exogenous repayment at rate λ, and has a marginal tax rate τ.

The optimal threshold from Agarwal et al. (2013) Theorem 2 is:

$$x^* = \frac{1}{\psi}\left[\phi + W\left(-e^{-\phi}\right)\right], \tag{5}$$

where $W(\cdot)$ is the Lambert-W function and:

$$\begin{aligned}
\psi &= \frac{\sqrt{2(\rho+\lambda)}}{\sigma}, \\
\phi &= 1 + \psi(\rho+\lambda)\frac{C(M)}{M}, \\
\frac{C(M)}{M} &= \frac{\kappa/M}{1-\tau}.
\end{aligned}$$

The exogenous repayment rate λ is calibrated following Appendix C of Agarwal et al. (2013):

$$\lambda = \mu + \frac{i_0}{e^{i_0\Gamma} - 1} + \pi,$$

where μ is the annual moving probability, i_0 is the original nominal rate, Γ is the remaining term, and π is inflation.

The tax-adjusted closing cost κ follows Appendix A of Agarwal et al. (2013):

$$\kappa = F + fM\left(1 - \frac{\tau}{\theta + r_d}\left[\frac{1 - e^{-(\theta + r_d)T_m}}{T_m} \cdot \frac{r_d}{\theta + r_d} + \theta\right]\right),$$

where F is the fixed closing cost, f is the points fraction, $\theta = \mu + 0.10$, $r_d = \rho + \pi$, and T_m is the mortgage term in years.

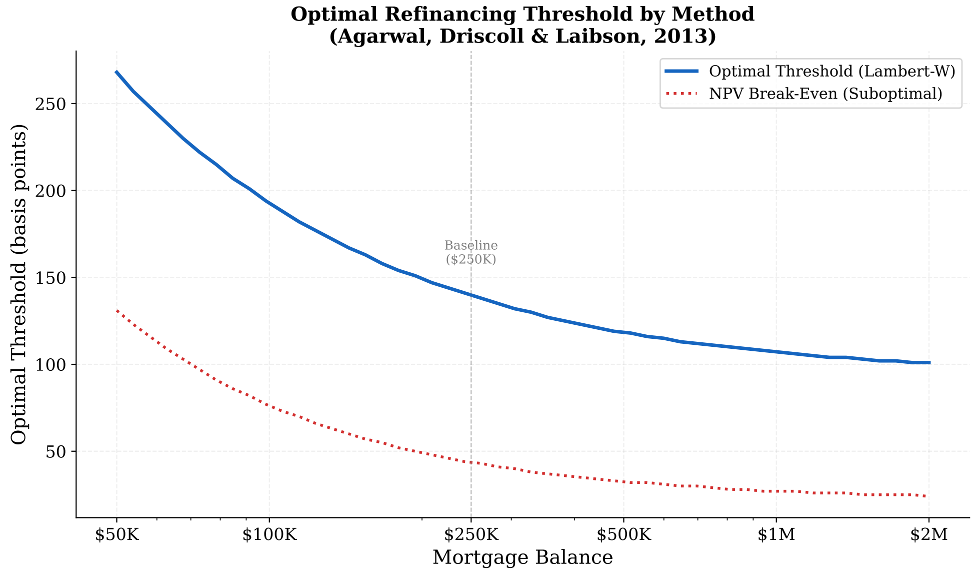
Using baseline parameters, the exact threshold reproduces Table 1 of Agarwal et al. (2013) (Appendix Table 6):

Appendix Table 6: Replication of Agarwal et al. (2013) Table 1: Optimal Refinancing Thresholds

Balance	Exact (bps)	NPV (bps)	Paper (bps)
\$100,000	193	76	193
\$250,000	139	44	139
\$500,000	118	33	118
\$1,000,000	107	27	107

Notes: Exact = Lambert-W closed-form threshold (Equation (5)). NPV = naive break-even threshold ignoring the option value of waiting, computed using the same tax-adjusted closing costs as the exact formula. Paper = exact threshold values reported in Table 1 of Agarwal et al. (2013). All thresholds in basis points; baseline parameters: $\rho = 0.05$, $\sigma = 0.0109$, $\tau = 0.28$, $\mu = 0.10$, closing costs = 1% + \$2,000.

Appendix Figure 5 illustrates how both thresholds vary with mortgage balance. The gap between the exact threshold and the NPV break-even is the "option value of waiting": for a \$250,000 mortgage, the exact threshold is 139 bps versus an NPV break-even of 44 bps—more than three times the naive level. Both thresholds decrease with mortgage size because fixed closing costs become proportionally smaller relative to the balance. With the January 2024 time-varying calibration (Section 2.5), the threshold falls to 91 bps because the larger balance and lower origination costs reduce per-dollar closing costs; setting $\Gamma = 30$ (instead of 25) has negligible effect on option values, and setting $\tau = 0$ (a non-itemizer) modestly increases the option value.



Appendix Figure 5: The exact optimal threshold (blue solid) is always well above the NPV break-even (red dotted) because the optimal rule accounts for the option value of waiting. Both thresholds decrease with mortgage size because fixed costs become proportionally smaller. Baseline parameters as in text.

C Term-Structure and Spread Dynamics

This appendix states the interest-rate and mortgage-spread model summarized in Section 2.4.

I model the short rate using the Hull–White one-factor model (Hull and White, 1990):

$$dr(t) = [\theta(t) - \alpha\, r(t)]\, dt + \sigma_{\mathrm{HW}}\, dW(t), \tag{6}$$

where α is the mean-reversion speed, σ_{HW} is the short-rate volatility, and $\theta(t)$ is calibrated analytically to fit the initial Treasury yield curve. This ensures that the model prices all observed bonds correctly at time zero, so simulated rate paths are consistent with the term structure observed at each origination date.

Mortgage rates along each simulated path are derived from model-implied 10-year Treasury yields using the Hull–White analytical bond pricing formula (Appendix E). At each time step, I compute the model-implied 10-year zero-coupon bond price $P(t, t+10)$ and invert it to obtain the yield $y(t, 10) = -\frac{1}{10} \ln P(t, t+10)$. This approach captures the full term-structure dynamics—including mean reversion, forward-rate drift, and volatility effects—rather than relying on a crude "short rate plus constant" approximation.

Mortgage rates do not move one-for-one with Treasury yields. Empirically, regressing monthly changes in the 30-year PMMS rate on changes in the 10-year Treasury yield (post-1990, expanding window) yields a pass-through coefficient $\beta \approx 0.60$, indicating that a 100 bps decline in Treasury yields translates into roughly a 60 bps decline in mortgage rates. Part of this attenuation is mechanical survey smoothing; an error-correction specification that imposes the long-run one-for-one relationship between the two levels is included with the spread-model robustness checks (Appendix Table 12). The residual—the mortgage–Treasury spread—is persistent and stochastic. I model it as an AR(1) process estimated from the time series of $u_t \equiv \mathrm{PMMS}_t - \beta \cdot \mathrm{DGS10}_t$:

$$u_t = c + \varphi\, u_{t-1} + \sigma_\varepsilon\, \eta_t, \qquad \eta_t \sim N(0,1), \tag{7}$$

with $\varphi = 0.97$ held fixed and the innovation standard deviation σ_ε vol-targeted: it is calibrated each month so that the total simulated mortgage-rate volatility matches the observed trailing 36-month PMMS volatility (Appendix K.1). The vol-targeted σ_ε is close to the legacy expanding-window estimate of roughly 16 bps per month in most periods, diverging mainly in the 2020s, when the $\beta < 1$ pass-through compressed the Treasury-driven component of mortgage-rate movements. The intercept c ensures mean reversion toward the historical average spread rather than anchoring permanently at the origination-date level.

The mortgage rate along each simulated path is then:

$$m(t) = m(0) + \beta\big[y(t,10) - y(0,10)\big] + (u_t - u_0), \tag{8}$$

where $u_0 = m(0) - \beta \cdot y(0, 10)$ is the initial spread residual. This formulation anchors the mortgage rate path at the observed contract rate $m(0)$, dampens Treasury rate movements by $\beta < 1$, and allows the spread to fluctuate around its long-run mean. The short-rate volatility σ_{HW} is calibrated to the 10-year Treasury yield volatility (not total mortgage rate volatility), so that the Hull–White model captures the pure interest rate component; the remaining spread variability is absorbed by the AR(1) process.

For each month in the sample, the model is calibrated to the observed yield curve (11 Treasury maturities fit with a cubic spline), with $\alpha = 0.05$ and short-rate volatility σ_{HW} calibrated analytically so that the model-implied volatility of 10-year Treasury yields matches the observed trailing 36-month estimate. Full calibration details follow in Appendix D; Appendix K confirms that the mean simulated path tracks the forward curve implied by the initial term structure.

The Hull–White drift is calibrated to the observed forward curve, while the Agarwal et al. (2013) threshold and the discounting use the observable real 10-year Treasury yield ρ_t in place of a time-preference parameter; this approach is standard in borrower-side mortgage valuation (Stanton, 1995), with the interest-rate dynamics consistent with the term structure at origination and the discounting reflecting the market opportunity cost of funds rather than an unobserved preference.

D Hull–White Calibration

The time-dependent drift $\theta(t)$ in Equation (6) is:

$$\theta(t) = \frac{\partial f}{\partial t}(0, t) + \alpha\, f(0, t) + \frac{\sigma_{\mathrm{HW}}^2}{2\alpha}\left(1 - e^{-2\alpha t}\right),$$

with $f(0, t)$ the instantaneous forward rate derived from observed Treasury yields.

For each month, the yield curve is fit to observed Treasury rates at up to 11 maturities using a natural cubic spline. The instantaneous forward rate is computed by numerical differentiation: $f(0, T) \approx [(T + \Delta T) \cdot y(T + \Delta T) - T \cdot y(T)]/\Delta T$ with $\Delta T = 0.01$ years. The mean-reversion speed is fixed at $\alpha = 0.05$ (half-life of approximately 14 years). The short-rate volatility σ_{HW} is calibrated so that the model-implied annualized volatility of

10-year Treasury yields matches the observed trailing 36-month estimate $\hat{\sigma}_T$. Using the Hull–White bond-pricing formula, the annualized volatility of the model-implied 10-year yield is $\sigma_{\mathrm{HW}} \cdot B(10)/10$, where $B(T) = (1 - e^{-\alpha T})/\alpha$ (equal to $B(0,T)$ in the notation of Appendix E). Setting this equal to $\hat{\sigma}_T$ gives $\sigma_{\mathrm{HW}} = \hat{\sigma}_T \cdot 10/B(10)$; for $\alpha = 0.05$, the multiplier is $10/7.87 \approx 1.27$. Paths are generated using Euler discretization with monthly time steps, fully vectorized over 10,000 paths.

E Model-Implied 10-Year Yield

The Hull–White model admits closed-form zero-coupon bond prices. The price at time t of a bond maturing at T is:

$$P(t,T) = A(t,T)\exp\bigl[-B(t,T)\,r(t)\bigr],$$

where

$$\begin{aligned} B(t,T) &= \frac{1 - e^{-\alpha(T-t)}}{\alpha}, \\ \ln A(t,T) &= \ln \frac{P^M(0,T)}{P^M(0,t)} + B(t,T)\,f(0,t) - \frac{\sigma_{\mathrm{HW}}^2}{4\alpha}\bigl(1 - e^{-2\alpha t}\bigr)B(t,T)^2, \end{aligned}$$

with $P^M(0,T)$ the market-observed discount factor and $f(0,t)$ the instantaneous forward rate.

The model-implied 10-year yield at simulation step t is:

$$y(t,10) = -\tfrac{1}{10}\ln P(t,t+10).$$

The mortgage rate built from $y(t,10)$ via Equation (8) then reflects the full term-structure dynamics—mean reversion, forward-rate drift, and volatility—rather than relying on a crude "$r(t)$ plus constant" approximation.

F Simulation and the PV-Equivalence Condition

This appendix details the Monte Carlo simulation and the formal definition of the option-adjusted cost rate summarized in Section 2.4.

For each month in the sample, I simulate 10,000 monthly short-rate paths over a 30-year horizon, convert each to a mortgage rate path using the bond pricing formula, and apply the Agarwal et al. (2013) threshold at each monthly step. Rather than truncate at an arbitrary fixed horizon, I weight each month's cash flows by the probability that the mortgage is still outstanding—the survival function $S(t) = e^{-\mu_{\text{tot}} t}$ implied by the exogenous (non-refinancing) termination hazard μ_{tot}—so that the effective horizon is the borrower's expected loan life rather than a fixed cutoff. (In the discrete equation below, $S_k = S(k/12)$.) Because refinancing resets the rate but keeps the borrower in the mortgage, the relevant hazard is the rate at which the loan terminates without refinancing. The model-implied expected life rises from about 8.7 years in the early 1990s to 12.3 years in the 2020s as homeowner mobility falls (Appendix P, Appendix Figure 14), sitting between the seven-year MBS weighted-average-life convention and the sixteen-year tenure implied by mobility alone. Appendix H confirms that results are well converged by 10,000 paths (Monte Carlo standard error $\approx \pm 1$ bp). The sensitivity analysis (Appendix Figure 9) and Appendix Table 13 confirm that assuming a fixed seven-year life lowers the option value while a thirteen-year life raises it, with neither change altering the qualitative time-series pattern. When the borrower's current rate exceeds the market rate by at least x^*, she refinances: her rate resets to the market rate, she pays closing costs out of pocket, and the mortgage term resets to 30 years with a new amortization schedule.

For each simulated path, the borrower's actual monthly cash flows are computed: at origination (and after each refinance), the monthly payment is recomputed for a fresh 30-year amortization at the new rate on the outstanding balance, and closing costs are recorded as additional cash flows. For the cost-rate calculation, nominal mortgage cash flows are discounted at the observable nominal 10-year Treasury yield $r_d(t) = y_{10}^{\text{Tsy}}(t)$ prevailing at origination, in place of a subjective rate; the real discount rate entering the Agarwal et al. (2013) threshold is correspondingly the real 10-year Treasury yield $\rho_t = y_{10}^{\text{Tsy}}(t) - \mathbb{E}_t[\pi_{10}]$ (Cleveland Fed expected inflation), so that $\rho_t + \mathbb{E}_t[\pi_{10}] = r_d(t)$. The parameters π, τ, and μ continue to enter the Agarwal et al. threshold as in the refinancing rule, with ρ replaced by the observable ρ_t.

Formally, the option-adjusted cost rate c^* is the constant annual mortgage rate such that the survival-weighted present value of payments and termination payoffs on a hypothetical no-option mortgage at c^* equals the mean survival-weighted present value of actual payments, refinancing costs, and termination payoffs across all simulated paths:

$$\sum_{k=1}^{T} S_{k-1}\,\mathrm{PMT}_k(c^*)\,e^{-r_d k/12} + \sum_{k=1}^{T}(S_{k-1} - S_k)\,B_k(c^*)\,e^{-r_d k/12} + S_T\,B_T(c^*)\,e^{-r_d T/12}$$
$$= \mathbb{E}\left[\sum_{k=1}^{T} S_{k-1}\,\mathrm{CF}_k(\omega)\,e^{-r_d k/12} + \sum_{k=1}^{T}(S_{k-1} - S_k)\,B_k(\omega)\,e^{-r_d k/12} + S_T\,B_T(\omega)\,e^{-r_d T/12}\right], \tag{9}$$

where $\mathrm{PMT}_k(c)$ is the scheduled monthly payment under the counterfactual no-option mortgage, $B_k(\cdot)$ is the outstanding balance at month k, $\mathrm{CF}_k(\omega)$ is the borrower's actual cash flow at month k along path ω (including any refinancing costs), $S_k = e^{-\mu_{\text{tot}}\,k/12}$ is the probability that the loan is still outstanding after k months (the month-k payment is weighted by survival to the start of the month, and with probability $S_{k-1} - S_k$ the loan terminates and its balance is settled), T is the 30-year horizon in months, and $r_d = y_{10}^{\text{Tsy}}$ is the nominal Treasury discount. This is solved numerically via bisection; both sides are survival-weighted and discounted at the same observable nominal Treasury rate, making this a PV-equivalence condition rather than an IRR. The option-adjusted cost rate is the single fixed rate at which the borrower would be indifferent between (a) paying the contract rate $m(0)$ with the option to refinance optimally and (b) paying a constant rate c^* with no option. The option value is then $V = m(0) - c^*$, expressed in basis points. A simpler rate-average measure and the legacy payment-only PV equivalent are reported as appendix comparisons (Appendix G).

All expectations are conditioned on the current yield curve and volatility estimate. The

Hull–White model is calibrated to the observed term structure, so the option-adjusted cost rate reflects forward-curve-consistent rate dynamics conditioned on current bond prices.

G Rate-Average Comparison

The primary metric throughout the paper is the terminal-balance-adjusted PV-equivalent option-adjusted cost rate c^* (Equation (9)), which accounts for payments, refinancing costs, amortization resets, and the remaining balance at the horizon. As a simpler alternative, one can compute a *rate-average* measure: the discount-weighted average of the coupon rates actually paid along each path,

$$c^{\mathrm{RA}} = \frac{\mathbb{E}\left[\int_0^H m_s(\omega)\, e^{-\rho s}\, ds\right]}{\int_0^H e^{-\rho s}\, ds},$$

where $m_s(\omega)$ is the mortgage rate paid at time s along path ω and ρ is the borrower's discount rate (a fixed rate here, in contrast to the observable ρ_t of the headline configuration). This rate-average measure captures the direct interest-rate savings from refinancing; closing costs enter indirectly through the optimal threshold but are not included as separate cash flows.

Across the full 434-month sample, the rate-average and terminal-balance-adjusted PV measures are highly correlated, but their levels can differ materially. In high-option-value scenarios the gap can be larger: for the January 2024 scenario, the rate-average measure yields an option value of 186 bps while the terminal-balance-adjusted PV-equivalent yields 133 bps. This divergence arises because each refinancing resets the amortization to a fresh 30-year schedule, temporarily increasing the interest share of early payments. The rate-average measure, which simply weights the rate paid at each point in time by the discount factor, does not capture this mechanical effect. The PV-equivalent is preferred because it accounts for the full cash-flow consequences of refinancing, including closing costs, amortization resets, and the remaining loan balance at the horizon.

H Monte Carlo Convergence

To verify convergence, I ran the simulation for October 2023 (contract rate 7.31%) with varying path counts (Appendix Table 7):

Appendix Table 7: Monte Carlo Convergence

Paths	Mean Effective Rate (%)	Option Value (bps)
1,000	6.41	90
5,000	6.42	89
10,000	6.42	89
50,000	6.42	89

Notes: Simulation uses the October 2023 yield curve (contract rate 7.31%). Mean effective rate is the terminal-balance-adjusted PV-equivalent cost rate. Option value = contract rate minus mean effective rate, in basis points. Fixed borrower parameters as in Appendix J ("Fixed" column).

Results are well converged by 10,000 paths, with the Monte Carlo standard error approximately ± 1 bps.

I Historical Borrower Calibration

This appendix documents the time-varying borrower calibration summarized in Section 2.5. The Agarwal et al. (2013) threshold depends on four borrower-side parameters—the mortgage balance M, the effective tax rate on mortgage interest τ, the annual moving probability μ, and the origination costs (points plus fixed costs)—that have changed substantially over the sample period. The baseline analysis calibrates each to publicly available time-varying data rather than fixing them at representative-borrower values, so that the threshold and option value reflect historical conditions at origination.

The average government-sponsored enterprise (GSE) loan balance—from the FHFA Monthly Interest Rate Survey spliced with the National Mortgage Database—rose from roughly \$102,000 in 1990 to \$350,000 in 2024. A larger balance reduces the per-dollar impact of fixed closing costs and lowers the refinancing threshold, making the option more valuable.

Origination fees and discount points are taken directly from the FRED series `MORTPTS30US`, which reports the combined fee charged by lenders as a percentage of the loan amount.[3] Points declined from about 2% in the early 1990s to 0.5–0.7% after 2005. Lower origination costs reduce the threshold and raise the option value. Because the survey contract rate is quoted conditional on these points, the initial points are already embedded in $m(0)$; points re-enter only as out-of-pocket costs at future refinances, so the option value measures the cost of exercising the option, not of originating the loan, and is distinct from the APR.

The annual moving probability μ is measured by the Census CPS/ASEC homeowner mobility rate: the fraction of persons in owner-occupied housing who report having moved in the preceding year.[4] Homeowner mobility fell from about 9% in the early 1990s to 4–5% in recent years; after adding the non-relocation hazard δ and survival-weighting the cash flows, this implies an increase in expected loan life from about 8.7 years in the early 1990s to 12.3 years in the 2020s (Appendix P, Appendix Figure 14). A lower μ raises the option value because a longer expected stay gives the borrower more time to encounter a favorable refinancing opportunity. The 2023 CPS rate of 4.1% partly reflects the "lock-in" of existing low-rate borrowers (Batzer et al., 2024); a new high-rate borrower who is not locked in would move sooner, shortening expected life toward the seven-year case in Appendix Table 13 and lowering the option value, so the headline, which uses the observed mobility, is if anything an upper bound for such a borrower.

The effective marginal tax rate on mortgage interest is computed as $\tau_{\text{eff}}(t) = \Pr(\text{itemize} \mid \text{mortgage}, t) \times \text{MTR}_{\text{itemizers}}(t)$, combining NBER TAXSIM marginal rates with IRS SOI

[3] This series is paired by construction with the PMMS: the reported contract rate is conditional on the borrower paying the reported points. The series was discontinued in November 2022 when Freddie Mac changed the PMMS methodology; post-2022 values are extrapolated at the 2022 average ($\approx$0.7%).

[4] The CPS series measures relocations directly; I add a constant $\delta = 0.025$ to capture the remaining non-refinancing terminations (in-place payoff, default, death) so that $\mu_{\text{tot}} = \mu_t + \delta$ is the full exogenous hazard of leaving the mortgage in the Agarwal et al. (2013) framework. This same hazard sets the survival-weighted horizon of Appendix F. See Molloy et al. (2011) for trends; Frost (2020) and Batzer et al. (2024) for recent declines.

itemization data. Before the Tax Cuts and Jobs Act, roughly 73–78% of mortgage holders itemized at marginal rates of 19–23% (Poterba and Sinai, 2008), yielding $\tau_{\text{eff}} \approx 0.15$–$0.18$. After the TCJA doubled the standard deduction in 2018, only about 27% itemize, yielding $\tau_{\text{eff}} \approx 0.065$. A lower tax rate reduces the tax benefit of mortgage interest and raises the effective cost of refinancing, modestly increasing the threshold.

At each future refinancing event within a simulation path, the borrower is assumed to face the same per-dollar origination cost and fixed closing cost ($2,000) as at the original origination. The discount rate is the observable real 10-year Treasury yield (Appendix F), time-varying over the sample; the remaining constants are inflation ($\pi = 0.03$, which enters only the threshold's tax term) and remaining term ($\Gamma = 25$). Appendix J reports the full parameter set, and Appendix Table 16 quantifies the contribution of each time-varying parameter.

J Baseline Parameters

Several exhibits report results under named model configurations. Appendix Table 8 defines each configuration once; table notes elsewhere refer to these names rather than re-describing parameters. Appendix Table 9 reports the baseline parameter values.

Appendix Table 8: Configuration glossary

Configuration	Definition
headline	Time-varying borrower calibration (balance, tax rate, mobility, and points from CPS/SCF/GSE series); observable real 10-year Treasury discount ρ_t; nominal-Treasury PV discount; survival-weighted horizon with hazard $\mu_t + \delta$ ($\delta = 0.025$); PV-equivalent measure; 10,000 paths; January 1990–February 2026. The configuration behind every headline number.
calibration_only	Time-varying borrower calibration as in *headline*, but with the legacy subjective discount $\rho = 0.05$, PV discount $\rho + \pi = 0.08$, and a fixed 10-year horizon. Isolates the calibration channel from the discount and horizon changes.
fixed_borrower	Legacy baseline: constant borrower parameters ($\rho = 0.05$, $\tau = 0.28$, $\mu = 0.10$, $\Gamma = 25$, \$250K balance), fixed 10-year horizon. Matches Agarwal et al. (2013)-style calibrations.
backtest_window	The *headline* configuration restricted to 1990–2016 origination cohorts with full 10-year realized windows ($N = 314$)—the cohorts for which the realized-path backtest is feasible.

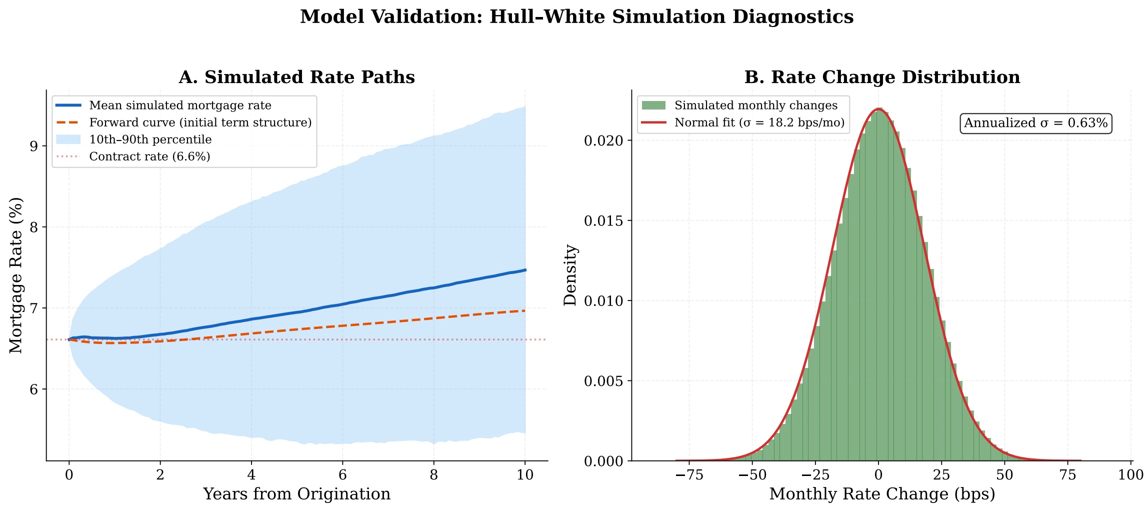
Appendix Table 9: Baseline Parameters

Parameter	Description	Fixed	Time-varying
ρ_t	Real 10-year Treasury yield	—	−0.8–4.9%
δ	Non-relocation hazard add-on	0.025	—
τ	Effective tax rate	0.28	0.065–0.18
μ	Annual moving probability	0.10	0.04–0.09
π	Inflation rate	0.03	—
Γ	Remaining mortgage term (years)	25	—
i_0	Original nominal rate	Contract rate	
M	Mortgage balance	\$250K	\$102K–\$365K
f	Refinancing points	1%	0.5–2.0%
F	Fixed closing cost	\$2,000	—
α	Hull–White mean reversion	0.05	—
N_{sim}	Simulation paths	10,000	
Horizon	Survival-weighted expected life	8.3–13.1 years (output)	

Notes: The "Fixed" column shows values used for the Agarwal et al. (2013) replication (Appendix B) and the sensitivity analysis (Appendix Figure 9). The "Time-varying" column shows the range of values used in the main historical analysis, calibrated to data as described in Section 2.5. A "—" indicates that the parameter is held constant. Sensitivity to each parameter is examined in Appendix L.1.

K Model Validation

Appendix Figure 6 presents two diagnostic checks on the Hull–White simulation.

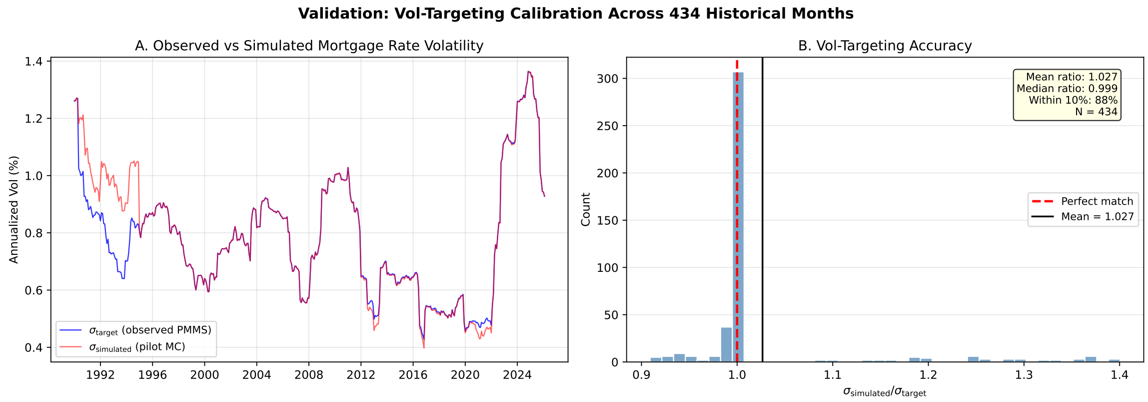


Appendix Figure 6: *Left:* Mean simulated mortgage rate path (blue) with the 10th–90th percentile fan, anchored at the January 2024 contract rate (6.61%). The dashed orange line shows the forward 10-year mortgage rate implied by the initial term structure, computed as $-\frac{1}{10}\ln[P(0,t{+}10)/P(0,t)] + s$. The mean simulated path tracks the forward curve closely, confirming correct calibration. *Right:* Distribution of simulated monthly mortgage rate changes across all 10,000 paths over a 10-year horizon. The histogram closely matches a normal distribution, consistent with the Gaussian diffusion assumption.

The left panel confirms that the mean simulated path is anchored at the observed contract rate, tracks the forward curve implied by the initial term structure, and exhibits gradual mean reversion toward the long-run rate level. The 10th–90th percentile fan illustrates the substantial uncertainty in rate paths over a 10-year horizon. The right panel shows that the distribution of monthly rate changes is approximately Gaussian, consistent with the Hull–White diffusion model. The annualized volatility of simulated changes is reported directly on the figure, providing a check against the target volatility from the rolling 36-month estimate.

K.1 Volatility Calibration Validation

The vol-targeting procedure calibrates the AR(1) spread innovation variance so that the total simulated mortgage rate volatility matches the observed trailing 36-month PMMS volatility. Appendix Figure 7 validates this calibration across all 434 historical months by simulating 5,000 pilot paths per month, discarding the first 36 months of each path as AR(1) burn-in, and comparing the annualized standard deviation of simulated monthly mortgage rate changes to the target.

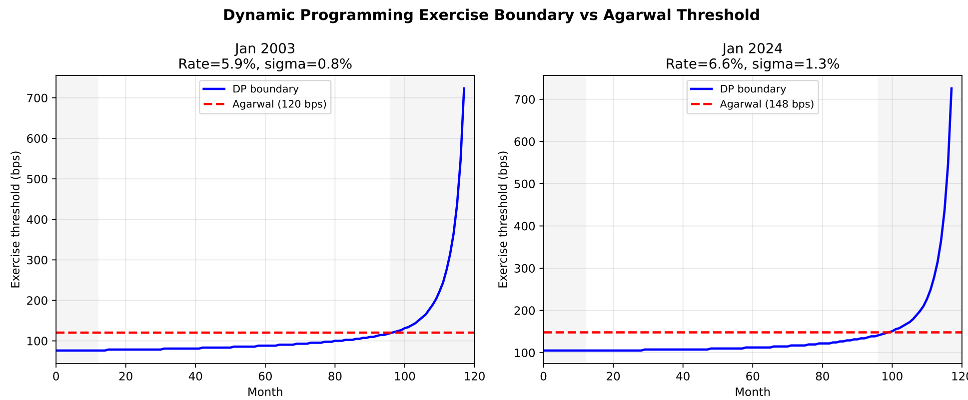


Appendix Figure 7: *Left:* Time series of target mortgage rate volatility (trailing 36-month PMMS) versus simulated mortgage rate volatility from 5,000 pilot paths. *Right:* Distribution of the ratio $\sigma_{\text{sim}}/\sigma_{\text{target}}$ across all 434 months. Median ratio is 1.00; 88% of months fall within 10% of unity.

The median ratio of simulated to target volatility is 1.00, with 88% of months falling within 10% of the target. Post-2000, the calibration is essentially exact (decade mean ratios of 1.00, 0.99, and 0.98 for the 2000s, 2010s, and 2020s respectively). The 1990s show modest over-simulation (mean ratio 1.12) because the expanding-window pass-through estimate $\hat{\beta}$ equals 1.0 in the earliest months when insufficient data are available to estimate $\beta < 1$; with a full pass-through there is no spread component to vol-target. Forcing $\beta = 0.60$ for the first 36 months (consistent with the full-sample estimate) lowers the 1990s decade mean option value by 14 bps but changes the full-sample mean by less than 4 bps.

K.2 Optimal Threshold Verification

The Agarwal et al. (2013) closed-form threshold is a constant approximation to the time-varying boundary that solves the borrower's optimal stopping problem. To verify the approximation, I solve the discrete dynamic programming (DP) problem on a grid of 500 rate-drop values (−400 to +800 bps) over 120 monthly time steps using backward induction with a Gaussian transition matrix.



Appendix Figure 8: DP exercise boundary (shaded, time-varying) versus Agarwal constant threshold (dashed) for two representative dates. The DP boundary is lower in the interior (months 12–96) and rises near the terminal date as the remaining horizon shrinks.

In the interior of the horizon (away from endpoints), the DP boundary is approximately 30–35 bps below the Agarwal threshold—meaning the Agarwal rule is conservative, triggering refinancing slightly later than optimal. Near the terminal date the DP boundary rises as the shrinking horizon reduces the option value of waiting, producing the "squeeze" visible in Appendix Figure 8. Using a constant threshold equal to the DP time-averaged boundary increases option value by approximately 2 bps for January 2003 (low-rate environment) and 5 bps for January 2024 (high-rate environment). The Agarwal closed-form threshold is therefore a reasonable approximation that slightly understates the option value.

K.3 Alternative Rate Model: CIR++

The Hull–White model allows negative interest rates, which could in principle distort simulated rate dynamics and the option value, especially when rates are high. To assess sensitivity to the rate process, I implement a CIR++ (shifted Cox–Ingersoll–Ross) alternative with closed-form bond pricing and a deterministic shift function $\xi(t)$ that matches the observed yield curve exactly; CIR++ enforces positive short rates and a square-root volatility structure. I recompute the option value for every month of the sample at the headline calibration, feeding CIR++-generated rate paths through the same simulation engine, survival-weighted horizon, and discounting used for the headline series, so that the two models differ only in the rate process. Appendix Table 10 summarizes the comparison.

Appendix Table 10: Hull–White vs. CIR++ option values, full sample

Statistic (434 months, headline calibration)	Value
Correlation of the HW and CIR++ option-value series	0.997
Mean absolute monthly difference	2.9 bps
Largest single-month difference	15.7 bps
Mean 2022–24 difference ($\lvert$CIR++ $-$ HW$\rvert$)	1.5 bps

Notes: CIR++ supplies the model-implied rate paths, which are then run through the same Monte Carlo engine, calibration, survival-weighted horizon, and discounting as the Hull–White headline, so the two models differ only in the rate process; the Hull–White column reproduces the headline series month for month. Option value is the terminal-balance-adjusted PV measure, 10,000 paths per month, 1990–2026. CIR++ enforces positive short rates. Generated by `12_cir_alternative.py -full`.

Across all 434 months the two option-value series are nearly identical: they correlate 0.997, with a mean absolute monthly difference of 2.9 bps. CIR++, which rules out the negative rates the Gaussian model permits, produces marginally *lower* values—by 1.5 bps on average over 2022–2024 and only a few basis points even in the highest-rate months—so the Gaussian rate assumption does not inflate the headline, and any difference is well inside the spread-model uncertainty of Section 3.4. The choice of rate process is therefore immaterial for the headline series.

K.4 Mechanical and Discretization Checks

Three discretization choices are verified via spot checks at representative dates. Replacing the 10-year Treasury (DGS10) with the 7-year (DGS7) as the rate anchor changes option values by 0.0 bps across three test dates—the yield curve fitting interpolates identically. Running with weekly time steps ($dt = 1/52$) instead of monthly ($dt = 1/12$) changes option values by at most 1.6 bps, confirming that the monthly discretization is adequate. Using 24-month or 60-month trailing volatility windows instead of the 36-month baseline shifts option values by ± 5 bps at individual dates without changing the qualitative pattern.

L Robustness

This appendix collects the paper's robustness evidence in one place: one-at-a-time parameter sensitivity, the spread-model specification, and the discount-rate and horizon assumptions. Appendix Table 11 is the summary index; the rate-model (CIR++), threshold-verification, volatility, and discretization checks it references live in the validation appendix (Appendix K).

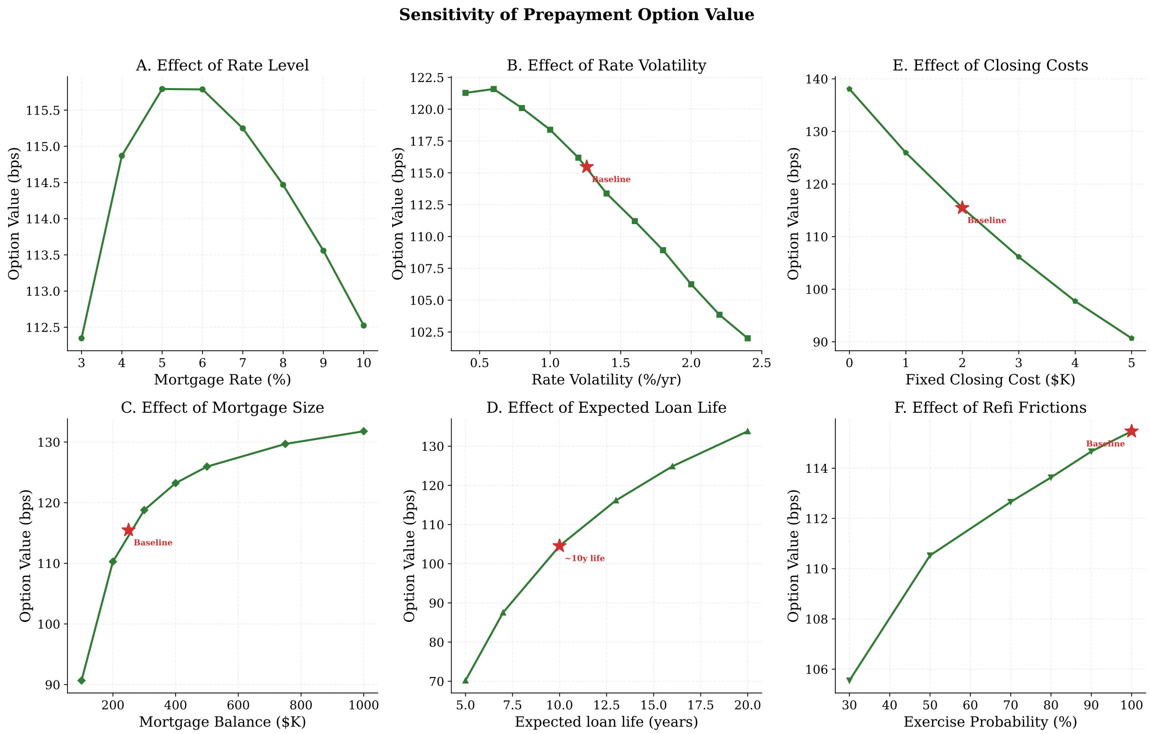
Appendix Table 11: Other Robustness Checks

Concern	Test (Location)	Impact
Gaussian rate model	CIR++ alternative (App. K.3)	Mean $\lvert\Delta\rvert = 2.9$ bps
Threshold accuracy	Discrete DP verification (App. K.2)	OV understated 2–5 bps
Vol calibration	Sim. vs realized vol (App. K.1)	Median ratio 1.00
Parameter sensitivity	10 variants at Jan 2024 (App. L.1)	±30 bps around baseline
Tenor and timestep	Spot checks (App. K.4)	<2 bps
Volatility window	24m vs 36m vs 60m (App. K.4)	±5 bps at individual dates

Notes: "Mean $\lvert\Delta\rvert$" is the mean absolute difference in option value between the alternative and baseline models. Full details in each referenced appendix.

L.1 Parameter Sensitivity

Appendix Figure 9 shows how the option value varies with six key parameters, each varied individually from baseline.



Appendix Figure 9: Panels B–F each vary one parameter while holding others at fixed baseline values (contract rate 6.61%, mortgage-rate volatility 1.26%, balance $250K, expected life 10 years, January 2024 yield curve). **A:** Rate level, using actual historical months (each point is one of the 434 origination months with its own yield curve and volatility). **B:** Rate volatility. **C:** Mortgage balance. **D:** Horizon. **E:** Closing costs. **F:** Exercise probability (refinancing frictions). Option value computed using the PV-equivalent measure (option-adjusted cost rate).

The option value increases with the rate level (Panel A), because a higher starting point provides more room for rates to fall. At 3%, the option is small (under 15 bps); at 7%+, it can exceed 100 bps. Rate volatility (Panel B) has the strongest effect, mirroring the standard result from option pricing that volatility benefits the option holder. Larger mortgage balances (Panel C) reduce the proportional impact of fixed closing costs, lowering the threshold and increasing the option value. Longer expected lives (Panel D) provide more time for favorable rate movements, though the marginal benefit diminishes as discounting and the falling survival probability reduce the weight on distant savings. Higher closing costs (Panel E) raise the refinancing threshold, reducing the option value by roughly 20–30% when costs are doubled.

Panel F introduces behavioral frictions by allowing the borrower to exercise with probability $p < 1$ when the threshold is met. Even modest frictions ($p = 0.5$) reduce the option value substantially, quantifying the cost of inertia documented by Keys et al. (2016).

A complementary check scales the analytically calibrated Hull–White volatility σ_{HW} by 0.7 and 1.5 at the January 2024 calibration: the option value shifts moderately in the expected direction, confirming that the volatility calibration matters quantitatively while the qualitative pattern—value increasing in volatility, rate level, and curve steepness—is robust.

L.2 Spread Model Specification

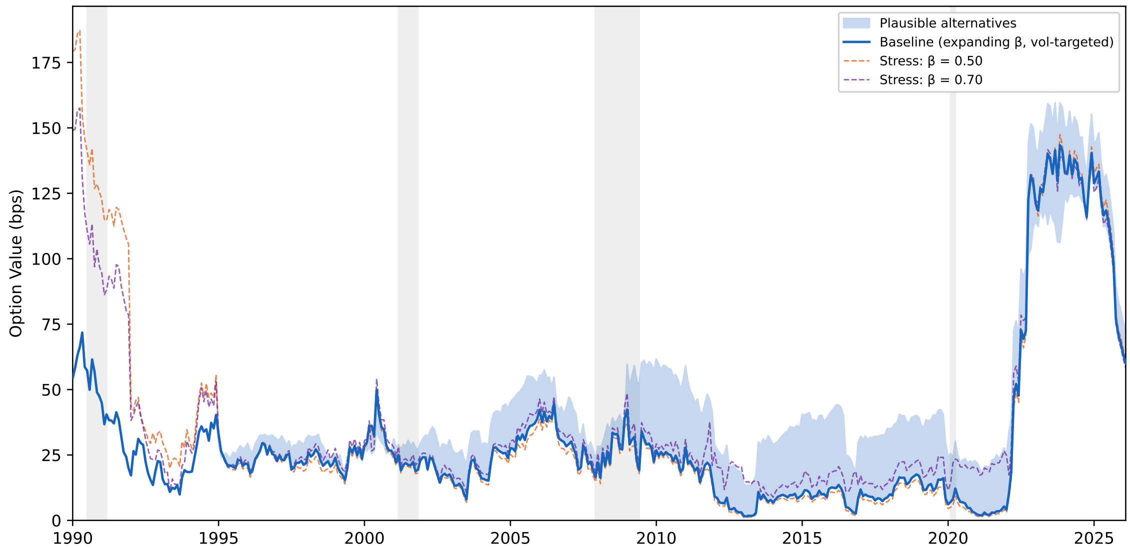
The baseline uses expanding-window $\hat{\beta}$ with vol-targeting because it (i) uses only information available at origination, (ii) matches realized mortgage-rate volatility by construction, and (iii) avoids parameter choices that are inconsistent with early-sample data (where the estimated pass-through is near 1.0). Appendix Table 12 reports the variants in two panels, each run across all 434 historical months with 10,000 paths per month: Panel A reruns every variant under the headline configuration (Appendix Table 8), so the robustness evidence covers the series the paper actually reports; Panel B preserves the calibration-only rows from earlier drafts for continuity. Under the headline configuration, Student-$t(5)$ innovations leave the mean essentially unchanged at 31 bps (vol-targeting pins the innovation variance, so tail shape barely matters); the random-walk and regime-switching variants raise it to 42 and 39 bps; and the error-correction variants—which address the downward bias in the monthly-changes pass-through directly—give 41 bps with the long-run coefficient estimated ($\hat{b} \approx 0.93$) and 48 bps with $b = 1$ imposed, with the divergence concentrated in the 2010s.

Appendix Table 12: Spread-model robustness (434 months, 10,000 paths each)

Specification	Mean	Median	Corr	1990s	2000s	2010s	2020s
Panel A: headline configuration							
Headline configuration (expanding β, vol-targeted)	31	22	1.000	28	26	12	74
Student-$t(5)$ innovations	31	22	1.000	28	26	12	74
Random-walk spread ($\varphi = 1$)	42	33	0.932	30	38	36	76
Regime-switching spread mean	39	28	0.958	28	27	33	89
Error-correction pass-through ($\hat{b}$ estimated)	41	31	0.954	34	35	25	91
Error-correction pass-through ($b = 1$ imposed)	48	38	0.922	33	46	35	96
Fixed $\beta = 0.50$ (stress)	36	22	0.864	48	25	11	74
Fixed $\beta = 0.70$ (stress)	38	25	0.914	43	29	18	79
Panel B: calibration-only configuration (for continuity with earlier drafts)							
Calibration-only baseline (legacy discount, fixed 10y)	25	18	1.000	25	21	8	59
Student-$t(5)$ innovations	27	20	0.905	42	20	5	48
Random-walk spread ($\varphi = 1$)	33	28	0.858	44	28	18	46
Regime-switching spread mean	32	22	0.937	42	20	17	59
Fixed $\beta = 0.50$ (stress)	35	18	0.581	75	18	4	48
Fixed $\beta = 0.70$ (stress)	35	22	0.649	66	22	8	51

Notes: Mean and median PV-equivalent option value in basis points; Corr = correlation with the panel's baseline series across 434 months; decade columns are decade means. Panel A runs every spread variant under the *headline* configuration (Appendix Table 8); its first row is the headline series itself. Panel B reruns the same variants under the *calibration_only* configuration (legacy subjective discount, fixed 10-year horizon), preserving the rows reported in earlier drafts. All runs use 10,000 Monte Carlo paths per month and the time-varying borrower calibration. Generated by analysis/make_spread_table.py.

Appendix Figure 10 displays the time series, with the shaded band spanning the plausible alternatives and the stress tests shown as dashed lines.



Appendix Figure 10: Baseline option value (solid blue) with shaded band spanning the four plausible spread specifications. Dashed lines show the two fixed-β stress tests. Recession periods shaded in gray.

Among the plausible alternatives, Student-$t(5)$ innovations leave the series essentially unchanged (correlation 1.000)—vol-targeting pins the innovation variance, so the option value is driven by variance rather than tail shape. The random-walk and regime-mean variants account for the upper end of the plausible range, with much of the divergence in the 2010s, where additional spread persistence or regime switching raises the decade mean from 12 to 33–36 bps. All plausible specifications preserve the broad time-series pattern (correlations of 0.92 or higher).

The fixed-β stress tests are deliberately extreme: they force a constant pass-through that is inconsistent with the early-sample data, where expanding OLS yields $\hat{\beta} \approx 1.0$. The result is a divergence concentrated in the 1990s, where the fixed $\beta < 1$ generates an artificially large spread component and raises the decade mean from 28 to 43–48 bps, but only modest differences post-2000. Even under these unfavorable assumptions the full-sample mean stays at 36–38 bps, within the range spanned by the plausible variants, though at the cost of a visibly distorted time path (correlations of 0.86–0.91).

L.3 Discount Rate and Horizon

The two assumptions replaced in this paper—the discount rate and the horizon—move the mean within a bounded range (Appendix Table 13). Holding the survival-weighted horizon fixed, the observable real-Treasury discount accounts for only about 2 bps; the horizon does the rest, with a fixed seven-year life giving 22 bps and a thirteen-year life 32 bps, bracketing the headline.

Appendix Table 13: Discount Rate and Horizon Robustness

Specification	PV-equiv OV (bps)	Rate-avg OV (bps)	Exp. life (years)
Baseline ($\rho = 5\%$, fixed 10y)	23	37	10.0
Empirical life, subjective discount	26	43	10.5
Fixed 7y life (MBS WAL)	22	42	6.9
Fixed 10y life	28	48	9.5
Headline (real disc., $\sim$12y)	**29**	**49**	10.5
Fixed 13y life	32	55	12.8
CPS mobility, no bump ($\sim$16y)	32	54	13.4

Notes: Mean option value (basis points) over 1990–2024 under alternative discount-rate and expected-life assumptions, varying one assumption at a time and holding the others at the headline settings. The headline uses the observable real 10-year Treasury discount and a survival-weighted horizon with hazard $\mu_t + \delta$ ($\delta = 0.025$). The full-sample (through February 2026) headline mean is 31 bps; this window is 1990–2024 for comparability across rows.

M Key Calibration Reference Values

Appendix Table 14 reports definitive results for three representative scenarios that appear throughout the paper. The high-rate scenario (6.61% contract rate on the January 2024 yield curve) is used in the worked example and sensitivity analysis. The low-rate scenario (January 2021) illustrates the small option value at cyclical rate troughs. The October 2023 scenario demonstrates behavior at the highest recently observed contract rates.

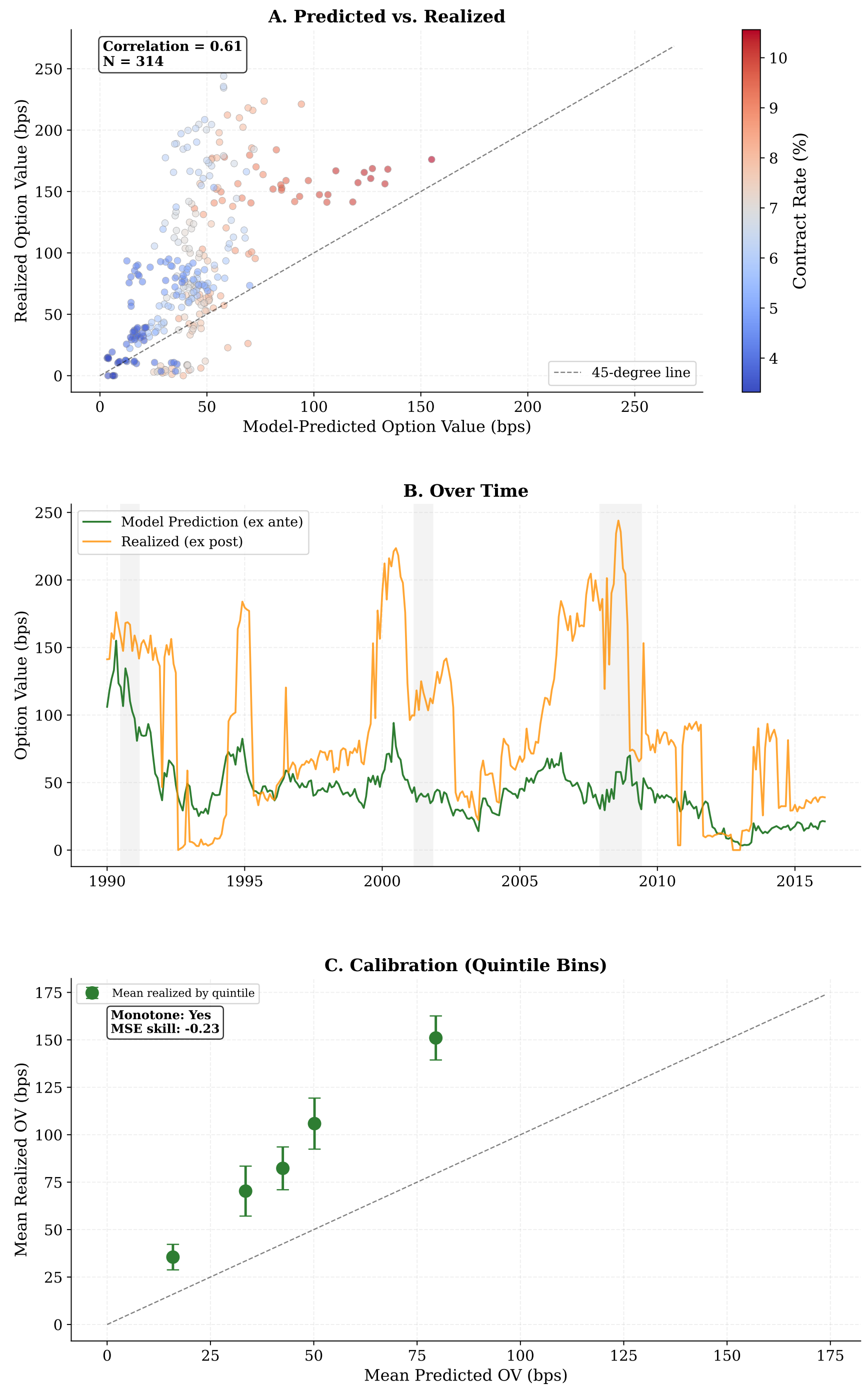
Appendix Table 14: Calibration Reference Values for Representative Scenarios

	6.61% (Jan 2024 YC)	2.67% (Jan 2021 YC)	7.31% (Oct 2023 YC)
Contract rate	6.61%	2.67%	7.31%
Balance	$345K	$288K	$343K
Eff. rate (rate-avg)	4.75%	2.62%	5.47%
Option value (rate-avg, bps)	186	5	184
Eff. rate (PV-equiv.)	5.28%	2.65%	6.01%
Option value (PV-equiv., bps)	133	2	130
Mean refinancings	2.80	0.17	3.00
Threshold (bps)	91	63	87
Paths	10,000	10,000	10,000

Notes: Each column uses the yield curve, contract rate, and time-varying borrower parameters (Section 2.5) observed on the indicated date. Eff. rate (rate-avg) = discount-weighted average coupon rate paid under optimal refinancing. Eff. rate (PV-equiv.) = constant mortgage rate whose discounted payments plus terminal balance, discounted at the nominal 10-year Treasury yield $r_d = y_{10}^{\text{Tsy}}$ (headline configuration; Appendix F), equal the mean discounted actual cash flows plus simulated terminal balance. Option value = contract rate minus effective rate, in basis points. Mean refinancings = average number of refinancing events per path. Threshold = Agarwal et al. (2013) optimal refinancing threshold for the indicated date's calibration.

N Historical Signal Content

A natural question is whether months with higher predicted option values also exhibit greater realized savings from refinancing—that is, whether the measure contains signal about future refinancing opportunities. Appendix Figure 11 compares the model's *ex ante* predicted option value (computed using only information available at origination) with the *ex post* realized option value (computed by applying the same optimal refinancing rule to the actual path of mortgage rates over the subsequent 10 years). The sample is restricted to 1990–2016 origination cohorts with full 10-year realized windows ($N = 314$)—the origination months for which a full 10-year realized window is available. Both axes use the rate-average measure: the realized value from a single observed path is a discounted average rate by construction, so the predicted value uses the matching rate-average column rather than the PV-equivalent measure of the headline series. This is the same cohort series as the surprise decomposition of Section 4, built from a single file (`surprise_by_cohort.csv`).

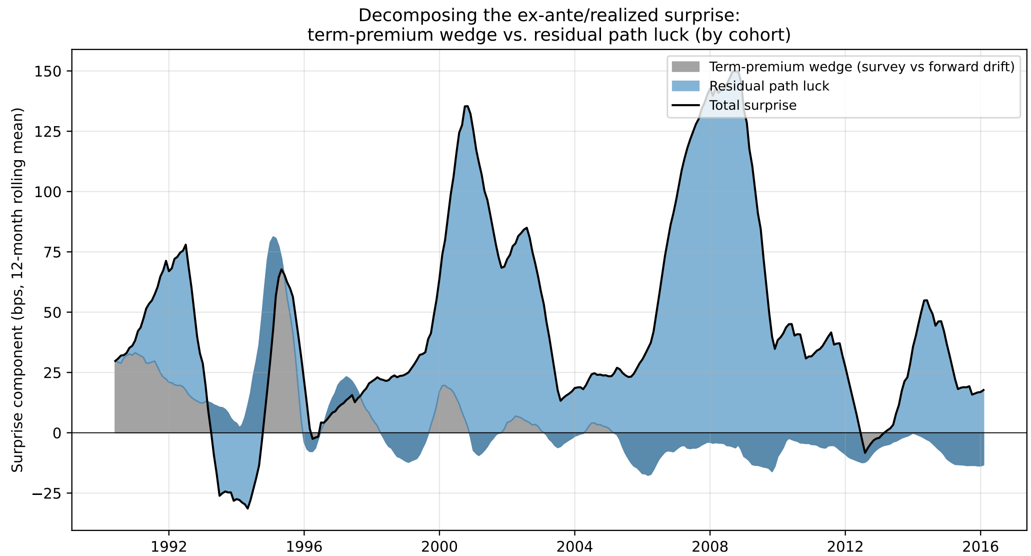


Appendix Figure 11: **A:** Each point is one origination month; x-axis is the model-predicted option value, y-axis is the realized value from actual rate paths. **B:** Time series of predicted versus realized values. **C:** Binned calibration plot—mean realized option value by quintile of predicted, with 95% confidence intervals. Sample restricted to 1990–2016 origination cohorts with full 10-year realized windows ($N = 314$) (originations through February 2016). Both axes use the rate-average measure, matching the realized single-path quantity; the series is identical to the surprise decomposition of Section 4 (source: `surprise_by_cohort.csv`).

The primary result is a correlation of 0.61 between predicted and realized option values: months with higher model-predicted option values do, in fact, experience significantly greater realized savings. The model systematically underpredicts—predicted mean 44 bps versus realized 89 bps, the same triple reported in Section 4—and the MSE skill score is negative (-0.23), meaning the model's point predictions are somewhat less accurate than simply predicting the sample mean. But point-prediction accuracy is not the relevant criterion; the question is whether the measure identifies high-versus-low option-value environments, which the correlation confirms.

The source of the systematic underprediction is economically intuitive. The backtest sample spans a period of extraordinary secular decline in interest rates—from roughly 10% in early 1990 to 2.65% in early 2021—punctuated by aggressive Federal Reserve interventions (the 2001 easing cycle, quantitative easing after 2008, the near-zero-rate era of 2020–2021). The Hull–White model, calibrated to the yield curve at each origination date, generates paths that mean-revert around the forward curve observed at that date. It cannot anticipate that rates will continue declining for decades, nor can it predict policy regime changes or financial crises. Consequently, a borrower originating at 8% in 1992 appeared to have modest option value according to the model (which expected rates to fluctuate near 8%), but in reality benefited from the secular decline to 5–6%—a windfall the forward curve never priced in. The pattern is starkest in the 1990s and 2000s, when realized refinancing opportunities were much larger than the forward-curve-consistent model values at origination.

The gap admits a formal decomposition. The Hull–White drift is fit to the forward curve, which embeds term premia; a borrower with survey-consistent expectations would have anticipated part of the realized decline. Rerunning the ex-ante value for every cohort under two alternative drift benchmarks—holding the volatility calibration and all random draws fixed—splits the 45-bp surprise into a predictable term-premium wedge and residual path luck: replacing the forward-curve drift with the SPF long-range 10-year Treasury expectation (linear convergence within a year; pre-1992 cohorts backfilled with the first 1992 survey) moves the mean ex-ante value by only 3 bps, and even a random-walk drift, which discards all forward-curve information, moves it by 16 bps—an upper bound on the predictable component. At least two-thirds of the surprise is residual path luck (42 bps). Appendix Figure 12 plots the split by cohort.



Appendix Figure 12: Decomposing the ex-ante/realized surprise into the term-premium wedge (survey-expectations vs forward-curve drift) and residual path luck, by origination cohort (1990–2016 origination cohorts with full 10-year realized windows ($N = 314$); rate-average measure; 12-month rolling means). The wedge is computed by rerunning the ex-ante value with the SPF long-range 10-year Treasury expectation in place of the forward-curve drift, holding the volatility calibration and random draws fixed; identity term premium + path luck = surprise holds cohort by cohort. Source: `surprise_decomposition.csv`.

Because the model value is computed using only information available at origination, the gap between the realized and predicted option values measures the *ex post* refinancing windfall generated by the realized interest-rate path. In the 1990–2016 backtest sample, realized values substantially exceed conditional model values not because the origination-date measure is the wrong *ex ante* measure, but because the subsequent period delivered a large downward rate surprise. Thus $V_t^{\text{realized}} - V_t^{\text{ex ante}}$ can be interpreted as a borrower-side option-value surprise: the component of refinancing value that was not priced by the origination-date yield curve. This quantity is distinct from the *ex ante* option-adjusted cost rate, but it is economically meaningful as a measure of the unexpected transfer to borrowers with callable fixed-rate debt, subject to the important caveat that actual realized gains depend on refinancing take-up, credit access, and borrower attention.

This limitation is inherent to any conditional model: the option values computed here reflect forward-curve-consistent rate dynamics at origination, not what subsequently transpired. A hypothetical sample with equal periods of rising and falling rates would exhibit less systematic bias. The relevant question for a prospective borrower is not "did the model predict the future correctly?" but rather "given what we know today about the yield curve and volatility, what is the expected benefit of the refinancing option?"—which is the conditional expectation the model provides.

O Borrower-Side Option Value and MBS Pricing

This appendix connects the borrower-side option value to the prepayment component priced in MBS markets: the construction of the spread series, one specification table containing every regression the paper runs, the time-series exhibits, and the interpretation caveats.

O.1 Construction

The total mortgage–Treasury spread—the gap between the primary mortgage rate (PMMS) and the 10-year Treasury yield—bundles secondary-market pricing of prepayment risk, originator margins and capacity constraints, credit risk, liquidity premia, and the effects of Federal Reserve MBS purchases (Boyarchenko et al., 2019). Because only the secondary-market component should directly reflect prepayment-option pricing, regressing the total spread on the option value is an apples-to-oranges test.

Following Fuster et al. (2013), I decompose the total spread into two components:

$$\underbrace{\text{PMMS} - \text{10y Treasury}}_{\text{Total spread}} = \underbrace{(\text{CC yield} - \text{10y Treasury})}_{\text{Secondary-market spread}} + \underbrace{(\text{PMMS} - \text{CC yield})}_{\text{Primary–secondary spread}}$$

where the current-coupon (CC) yield is the Fannie Mae 30-year MBS yield (Bloomberg:

MTGEFNCL). The secondary-market spread reflects compensation demanded by MBS investors for prepayment risk, duration, and convexity; the primary–secondary spread reflects originator profits, guarantee fees, and capacity constraints.

To construct the dependent variable, I decompose the MBS spread—the gap between the current-coupon Fannie Mae MBS pass-through yield (MTGEFNCL) and the 10-year Treasury—into the Bloomberg option-adjusted spread (OAS), which captures non-option factors, and the residual market option cost:[5]

$$\text{market option cost}(t) = \underbrace{(y_{\text{MBS}}(t) - y_{10}(t))}_{\text{MBS spread}} - \underbrace{\text{OAS}(t)}_{\text{non-option}}.$$

As a cross-check on the decomposition, the primary–secondary spread correlates 0.65 with the New York Fed's Originator Profit and Unmeasured Cost (OPUC) series over the 384 overlapping months.

O.2 Specifications

Appendix Table 15 reports every spread series regressed on the headline option value; each slope the paper has ever quoted appears here, once. The option value loads on the secondary-market spread (the preferred specification, slope 0.56, Newey–West $t = 4.94$, $R^2 = 0.23$) and not on the primary–secondary spread, consistent with investors pricing prepayment risk specifically; the OAS itself is a placebo and carries little option signal; and netting the OAS out of the secondary spread isolates a market option cost whose co-movement with the model is strongest of all—the same series used as the market cost basis in Appendix Table 1. Adding the VIX and changes in Federal Reserve MBS holdings as controls roughly doubles the explained variance while leaving the option-value slope unchanged, consistent with risk appetite and the Fed footprint as complementary spread drivers (Boyarchenko et al., 2019). The slope is not a deep behavioral parameter or a literal exercise probability: it blends

[5]The current-coupon yield is the pass-through rate to investors, net of the guarantee fee and servicing strip. It prices a hypothetical par-valued 30-year Fannie Mae TBA, not a portfolio of seasoned loans, making it directly comparable to the model's new-origination option value.

actual exercise behavior with differences in rate models, horizon definitions, and the gap between risk-neutral and physical measures.[6]

Appendix Table 15: Spread series regressed on the borrower-side option value

Dependent variable	Slope	NW t	R^2	N	Sample
Total spread (PMMS − 10y Treasury)	0.77	7.33	0.36	433	1990–2026
Secondary-market spread (CC − 10y Treasury)	0.56	4.94	0.23	433	1990–2026
Primary–secondary spread (PMMS − CC)	0.20	1.40	0.02	433	1990–2026
OAS (placebo: non-option factors)	-0.10	-0.86	0.01	353	1996–2026
Market option cost (secondary spread − OAS)	0.67	7.72	0.47	353	1996–2026
Secondary spread, VIX + ΔFed MBS controls	0.56	5.65	0.46	277	2003–2026

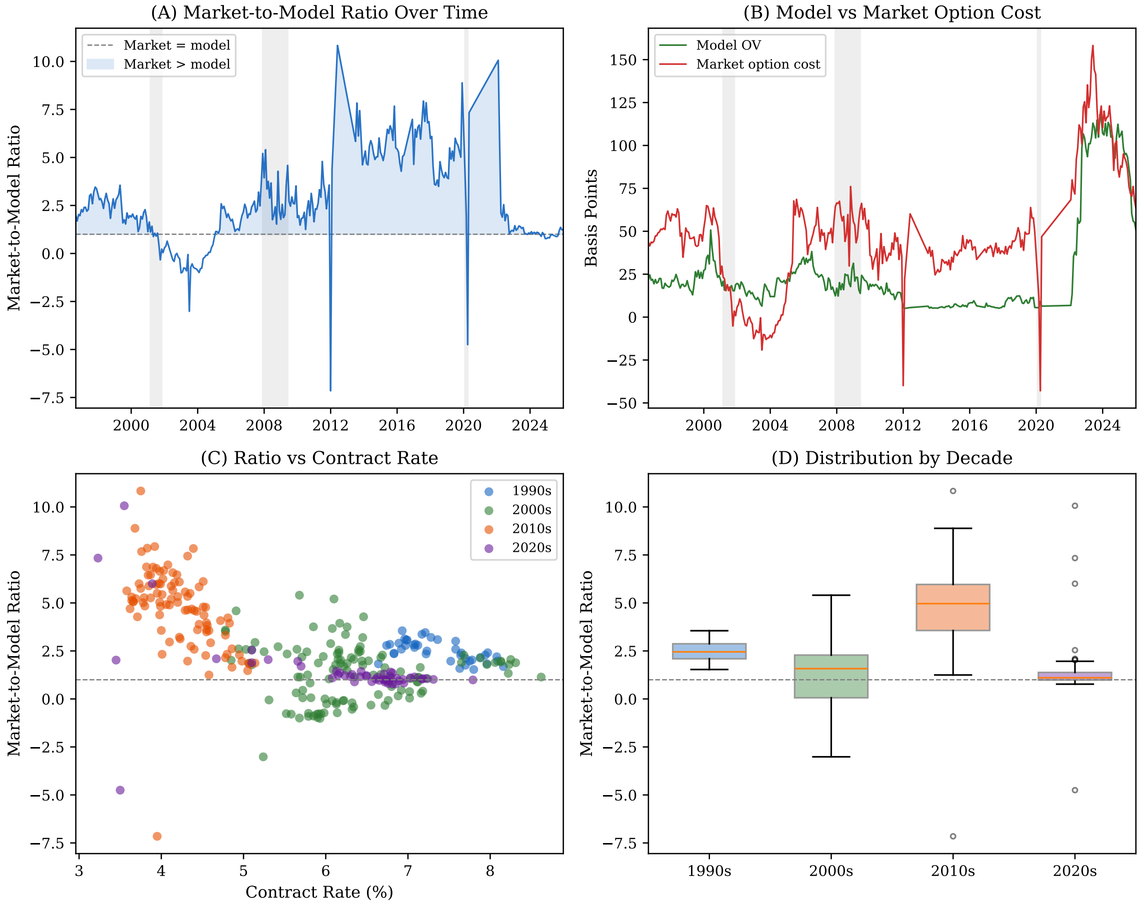
Notes: Monthly levels regressions of each spread series (bps) on the headline PV-equivalent option value (bps); Newey–West standard errors, 12 lags. The bold row is the preferred specification (the secondary-market spread is where prepayment risk is priced; its slope is the manifest value quoted in the text). The OAS row is a placebo: the option-adjusted spread should and does carry little option signal. The market-option-cost row (spread net of OAS) isolates the prepayment component and is the basis series used for the cost-basis analysis (Appendix Table 1); its sample starts with OAS coverage (September 1996). The controls row adds the VIX level and the monthly change in Federal Reserve MBS holdings. Generated by analysis/make_mbs_table.py.

O.3 Time-Series Exhibits

The preferred prepayment series—the market option cost against the model value, and the decomposition of the MBS spread into its OAS and option components—are plotted in Figure 2 in the main text. Appendix Figure 13 provides additional detail on the market-to-model ratio. Panel A plots the ratio of market option cost to the model option value over time. Panel B overlays the model option value and the market option cost, illustrating their co-movement. Panel C shows the ratio against the contract rate level. Panel D reports the

[6] The Bloomberg OAS uses a proprietary prepayment model under the risk-neutral measure; the model option value uses Hull–White dynamics under the physical measure. The slope therefore captures the net effect of suboptimal exercise, model differences, and any prepayment risk premium. The slope falls as the assumed expected life lengthens, because a longer life raises the model option value while the market cost is fixed; it remains positive and strongly significant across the expected-life range in Appendix Table 13.

distribution by decade.



Appendix Figure 13: Market-to-model option-cost ratio. Panel A: market option cost / model option value over time, with ratio = 1 line (dashed). Values above 1.0 (shaded) reflect periods where the market prices more prepayment risk than the model predicts. Panel B: model option value and market option cost time series. Panel C: ratio vs. contract rate, colored by decade. Panel D: box plots of the ratio by decade.

O.4 Interpretation and Caveats

The positive market relationship is consistent with independent evidence from the behavioral refinancing literature. Keys et al. (2016) find that many unconstrained borrowers who face a clear refinancing incentive nevertheless fail to exercise, while Gerardi et al. (2023) document racial disparities that imply lower effective exercise rates for minority borrowers, and Berger et al. (2024b) show that even small inattention frictions generate large welfare losses from delayed exercise. Those frictions help explain why a borrower-side optimal-exercise benchmark need not map one-for-one into MBS prices, even when both series reflect similar underlying economics.

The 26-basis-point intercept is a measurement artifact rather than an economic wedge. The market option cost is computed as the nominal spread to the 10-year Treasury minus the OAS, but the OAS is fitted to the full Treasury curve. The mismatch between a single-point benchmark and a curve-based computation generates a positive residual that varies with curve shape.[7]

Two further caveats. First, the model computes option values under physical-measure dynamics using the borrower's discount rate, whereas MBS investors price prepayment risk under the risk-neutral measure; the two need not coincide exactly—the gap is precisely the prepayment-risk-premium wedge documented in the cost-basis analysis (Appendix Table 1). Second, the 10-year Treasury is an imperfect duration benchmark for 30-year mortgages; a duration-matched benchmark would sharpen the decomposition and is left to future work.

P Supplementary Ex-Ante Exhibits

This appendix collects exhibits that support the ex-ante analysis of Section 3 but are not essential to the main narrative: the decomposition of the time-varying calibration and the expected-loan-life input series, additional views of the option-value distribution including the by-regime summary, sample simulated rate paths, and the full refinancing-event distribution.

P.1 Time-Varying Calibration Decomposition

Appendix Table 16 quantifies the contribution of each time-varying parameter by sequentially adding them to a fixed representative-borrower baseline ($250,000 balance, $\tau = 0.28$, $\mu = 0.10$, points = 1%). Holding the discount rate and horizon fixed to isolate the calibration channel, the full-sample mean is little changed by time-varying calibration (25 vs.

[7] Replacing the 10-year benchmark with the 7-year or 20-year Treasury shifts the intercept substantially (56 and −32 bps, respectively) while leaving the slope in the range 0.43–1.03, confirming that the intercept is benchmark-dependent but the positive marginal relationship is robust.

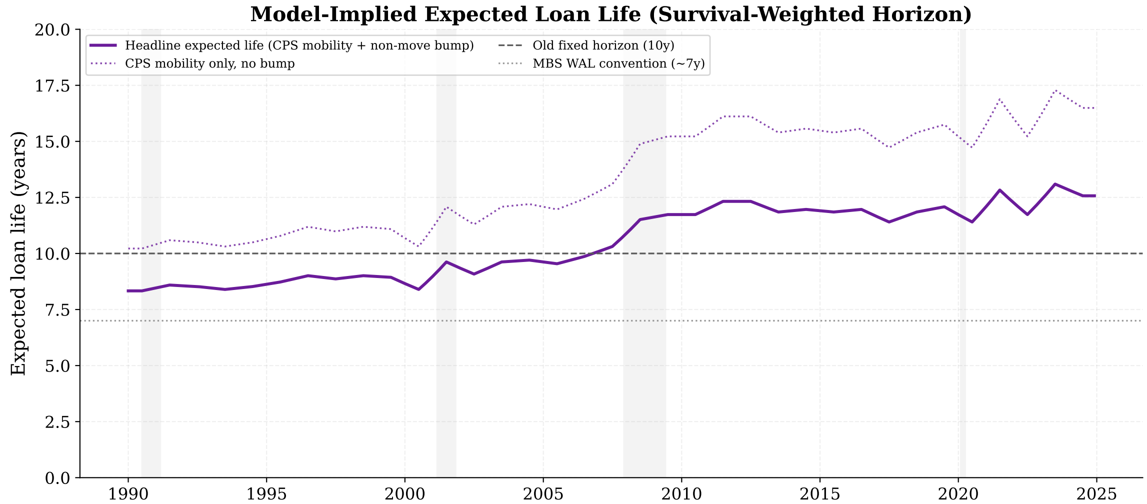
27 bps), though the time-series pattern shifts; the headline series of Section 3 additionally applies the observable real-Treasury discount and the survival-weighted horizon, which raise the mean to 31 bps. The 1990s option value falls from 42 to 25 bps, driven by smaller loan balances and higher origination costs: in the early 1990s, the average GSE loan was roughly $105,000 with 2% in points, making refinancing less cost-effective. The 2020s rise from 48 to 59 bps, reflecting lower origination costs and larger balances that together reduce the refinancing threshold. Mobility and the effective tax rate each contribute roughly 1 bp to the full-sample mean.

Appendix Table 16: Counterfactual Decomposition of Time-Varying Calibration

Specification	Full	1990s	2000s	2010s	2020s
Fixed baseline	27	42	20	5	48
+ Time-varying balance	23	29	16	5	52
+ Time-varying points	23	23	19	6	57
+ Time-varying mobility	24	24	20	7	58
+ Time-varying tax rate (= full)	25	25	21	8	59
+ Real-Treasury discount, survival horizon (= headline)	31	28	26	12	74

Notes: PV-equivalent option value in basis points (decade means). Each row adds one time-varying parameter while holding the others at their previous values. The fixed baseline uses $250K balance, $\tau = 0.28$, $\mu = 0.10$, and 1% origination points. The sequential ordering is balance, origination points, mobility, tax rate. All decomposition rows hold the discount rate and horizon at the fixed-baseline settings ($\rho = 5\%$, fixed 10-year horizon) and use 5,000 Monte Carlo paths per month across 434 historical months; the final row additionally applies the observable real-Treasury discount and the survival-weighted horizon (the *headline* configuration, 10,000 paths), landing on the headline mean.

Appendix Figure 14 plots the survival-weighted expected loan life implied by the mobility series—the input behind the horizon channel in the decomposition.



Appendix Figure 14: Model-implied expected loan life over time. The survival-weighted horizon uses the exogenous (non-refinancing) termination hazard $\mu_{\text{tot}} = \mu_t + \delta$, where μ_t is the CPS homeowner-mobility rate and $\delta = 0.025$ captures non-relocation terminations (in-place payoff, default, death). Expected life rises from about 8.7 years in the early 1990s to 12.3 years in the 2020s as mobility falls, sitting between the seven-year MBS weighted-average-life convention and the sixteen-year tenure implied by mobility alone.

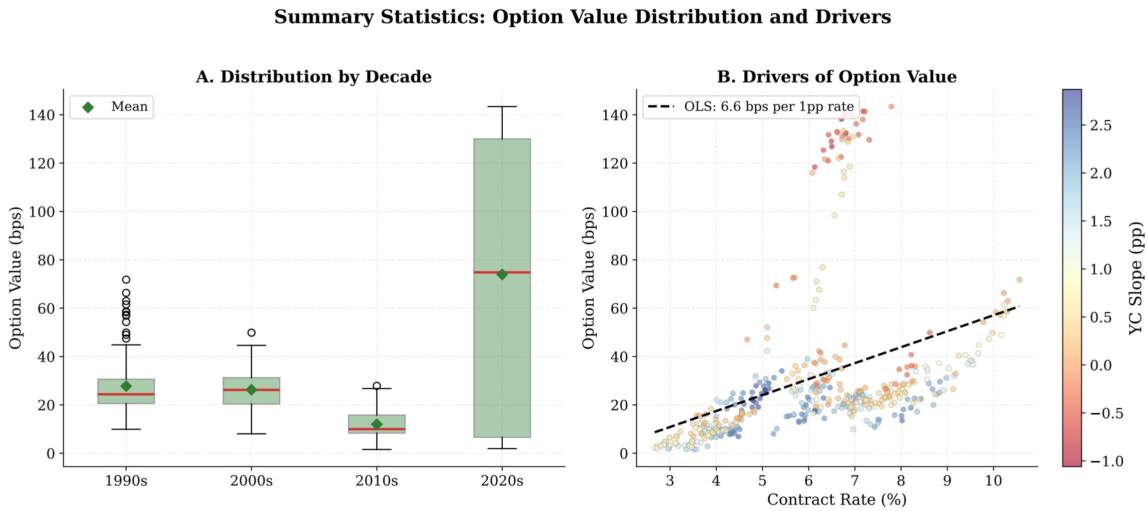
P.2 Additional Views of the Option-Value Distribution

Appendix Table 17 summarizes the key inputs and outputs by rate regime; Appendix Figures 15 and 16 show the option-value distribution and its drivers.

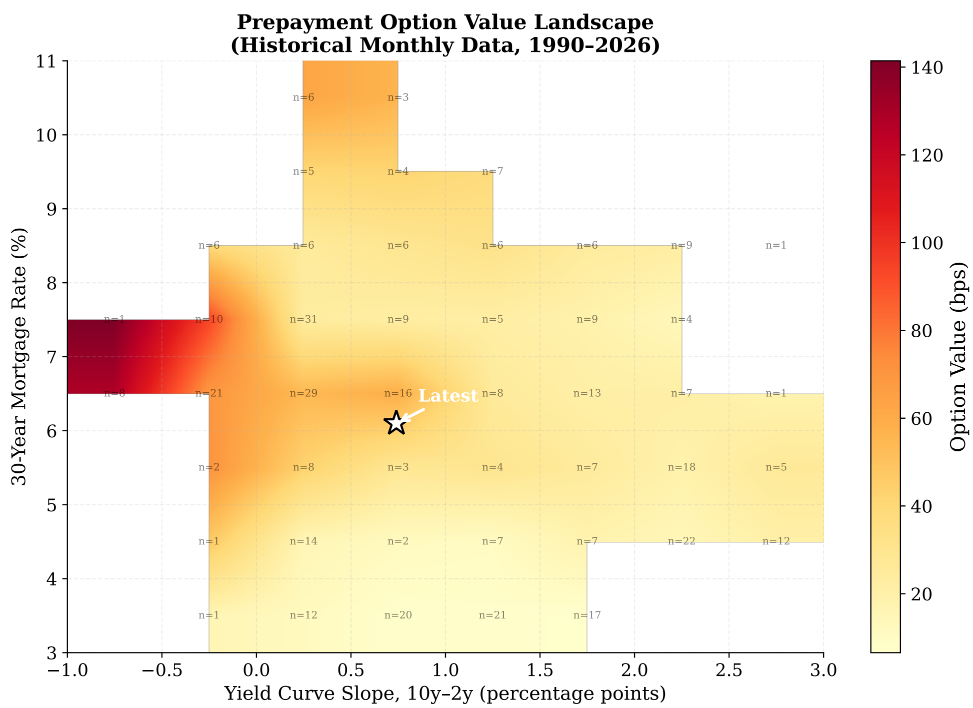
Appendix Table 17: Rate Environment and Option Value by Regime

	Low (<4%)	Medium (4–6%)	High (>6%)	2022–24
N (months)	83	113	238	36
Mean rate (%)	3.5	4.9	7.4	6.2
Mean slope (pp)	1.03	1.66	0.69	−0.27
Mean volatility (%)	0.55	0.78	0.85	1.07
Mean spread (bps)	182	174	180	260
Mean threshold (bps)	68	88	127	86
Mean refinancings	0.36	0.74	1.27	2.55
Mean OV (bps)	7	21	44	110
Median OV (bps)	7	18	27	130

Notes: Columns group months by the 30-year fixed mortgage contract rate at origination; the 2022–24 column is a separate grouping by date. Mean rate = average contract rate in the bin. Slope = 10-year minus 2-year Treasury yield (pp). Volatility = trailing 36-month annualized standard deviation of monthly mortgage rate changes. Spread = PMMS 30-year rate minus 10-year Treasury yield. Threshold = Agarwal et al. (2013) optimal refinancing threshold. Mean refinancings = average number of refinancing events per path over the loan's expected life. OV = option value ($m(0) - c^*$, basis points).



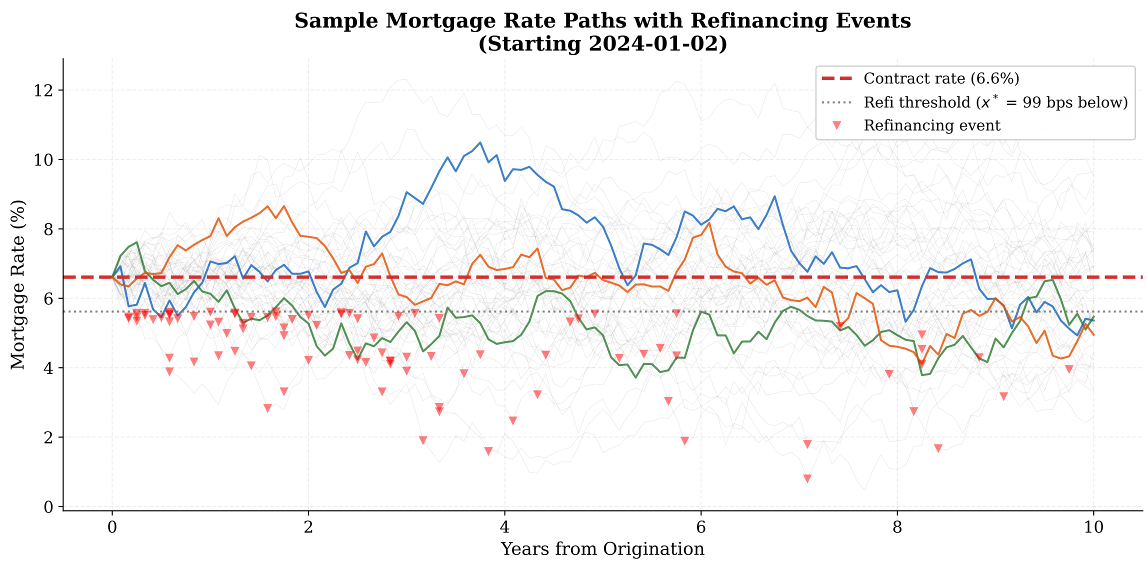
Appendix Figure 15: *Left:* Box plots of the option value by decade (median, IQR, 5th–95th whiskers). *Right:* Option value versus contract rate, colored by yield curve slope. Based on 434 monthly observations, 10,000 paths per month, survival-weighted horizon. Option value computed using the PV-equivalent measure (option-adjusted cost rate).



Appendix Figure 16: Each cell shows the mean option value (in bps) for months falling in that (mortgage rate, yield curve slope) bin. The star marks the most recent observation. Based on 434 monthly observations, January 1990 through February 2026. Option value computed using the PV-equivalent measure (option-adjusted cost rate).

P.3 Simulated Rate Paths

Appendix Figure 17 shows sample Monte Carlo paths for a borrower originating a mortgage in January 2024. Under the January 2024 calibration-only configuration ($345K balance, $\tau = 0.065$, $\mu = 0.043$, 0.7% points), the optimal threshold is 99 bps—well above the naive NPV break-even of roughly 30 bps. The optimal borrower waits for roughly three times the naive break-even, because exercising the option today forecloses the possibility of exercising at an even better rate tomorrow. The paths illustrate the heterogeneity of outcomes—some generate multiple refinancing events as rates follow a downward trajectory, while others never trigger a refinance because rates rise or remain near the initial level.



Appendix Figure 17: Each thin line is one simulated mortgage rate path over 10 years from the January 2024 yield curve (contract rate 6.61%). The dashed red line is the initial contract rate. The dotted gray line marks the optimal refinancing threshold (contract rate minus x^*). Red triangles mark refinancing events.

P.4 Refinancing-Event Distribution

Appendix Table 18: Refinancing Event Distribution by Scenario (10,000 paths, survival-weighted horizon)

	Jan 2021 (2.67%)	Jan 2024 (6.61%)	Oct 2023 (7.31%)
P(0 refis)	86%	3%	1%
P(1 refi)	11%	14%	10%
P(2 refis)	2%	26%	24%
P(3+ refis)	0%	57%	65%
Mean refis	0.17	2.81	3.04
Median refis	0	3	3

Notes: Each column is a representative origination month; the contract rate and yield curve are taken from that date. P(k refis) = fraction of 10,000 simulated paths in which the borrower refinances exactly k times (or ≥ 3 for the last row) over the loan's expected life, applying the Agarwal et al. (2013) optimal threshold at each monthly step. Borrower parameters are time-varying (Section 2.5); remaining parameters as in Appendix J.

Q Replication Materials

The ex-ante simulation is written in Python using NumPy, SciPy, and Pandas; interest-rate and mortgage data are downloaded from the FRED API at a fixed February 2026 vintage. Two discontinued or spliced series are handled explicitly: the points series `MORTPTS30US` (discontinued November 2022) is held at its 2022 average thereafter, and the average GSE balance splices the FHFA MIRS (through 2019) with the NMDB (2019 onward). The realized-gains pipeline additionally uses the HMDA Loan/Application Register files, the Freddie Mac single-family loan-level dataset and Quarterly Refinance Report, and the Survey of Consumer Finances; every headline table is regenerated from these public inputs by a single orchestrator script. The replication package is available at `https://www.adamlooney.com/data/mortgage-option-value`.